

EXACT REFORMULATIONS AND ROUTE AUDITS FOR DECIMAL DIGIT QUESTIONS ABOUT π : A TECHNICAL COMPANION

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ABSTRACT. This technical companion fixes the target specification and implication matrix for decimal digit questions about π . It proves three exact reformulations of the constant-word targets, with explicit scopes and hypotheses: the oriented decimal BBP condition, the Machin $3/7$ bracket condition, and the carry-certificate formulation. Across these routes it isolates a common location-and-carry obstruction, audits close-pair, valuation, positive-tail, Ramanujan, and Bailey–Crandall arguments, and states precisely which steps remain unproved. It also records property-admission results and the supporting R1, R3, and R4 lemmas, gives the complete open-problem ledger, and documents the machine-checked interface and its trust boundary. The note supplies exact reductions, conditional implications, separators, and reproducibility data; it does not prove decimal disjunctivity, normality, or either directed constant-word assertion for π .

1. TARGET SPECIFICATION FOR π

1.1. Canonical conventions. Put

$$\mathbb{N}_0 := \{0, 1, 2, \dots\}, \quad \mathbb{N}_+ := \{1, 2, 3, \dots\}, \quad \mathcal{D}_b := \{0, 1, \dots, b-1\}.$$

For $y \in \mathbb{R}$, write

$$\{y\} := y - \lfloor y \rfloor, \quad \|y\|_{\mathbb{R}/\mathbb{Z}} := \min_{m \in \mathbb{Z}} |y - m|.$$

A word of length k over $\mathcal{D}_b$ is an element of $\mathcal{D}_b^k$. Leading zeroes are significant and occurrences may overlap.

For every integer $b \geq 2$, use the unique base- b expansion that is not eventually $b-1$:

$$\{\alpha\} = \sum_{n=0}^{\infty} d_n^{(b)}(\alpha) b^{-(n+1)}, \quad d_n^{(b)}(\alpha) = \lfloor b^{n+1} \{\alpha\} \rfloor - b \lfloor b^n \{\alpha\} \rfloor.$$

Thus terminating expansions have trailing zeroes. Since π is irrational, it has no dual-expansion ambiguity in any integer base [19]. Its decimal digits are

$$d_n := d_n^{(10)}(\pi) = \lfloor 10^{n+1} \pi \rfloor \bmod 10, \quad n \in \mathbb{N}_0.$$

The indexing is zero-based after the decimal point; the integer part is not part of the stream.

For $w \in \mathcal{D}_b^k$, define

$$C_b(w, N; \alpha) := \#\{n \in \mathbb{N}_0 : n < N \wedge (\forall i \in \mathbb{N}_0) (i < k \Rightarrow d_{n+i}^{(b)}(\alpha) = w_i)\}.$$

Every start $n < N$ is counted, even if its word extends beyond position $N - 1$. For a nonempty word $w = w_0 \cdots w_{k-1}$, put

$$[w]_b := \sum_{i=0}^{k-1} w_i b^{k-1-i}, \quad I_b(w) := \left[\frac{[w]_b}{b^k}, \frac{[w]_b + 1}{b^k} \right).$$

With $T_b x = \{bx\}$, the word w occurs at position n exactly when $T_b^n \{\alpha\} \in I_b(w)$.

For the decimal orbit

$$x_n := \{10^n \pi\},$$

one has

$$x_n = \sum_{j=0}^{\infty} d_{n+j} 10^{-(j+1)}, \quad d_n = \lfloor 10x_n \rfloor, \quad x_{n+1} = 10x_n - d_n.$$

In particular, for $k \geq 1$,

$$\begin{aligned} d_n \cdots d_{n+k-1} = 0^k &\iff 0 \leq x_n < 10^{-k}, \\ d_n \cdots d_{n+k-1} = 9^k &\iff 1 - 10^{-k} \leq x_n < 1. \end{aligned}$$

1.2. The target nodes. The constant-word nodes are

CW0	$\iff$	$(\forall k \in \mathbb{N}_+)(\exists n \in \mathbb{N}_0)(\forall i < k) d_{n+i} = 0,$
CW9	$\iff$	$(\forall k \in \mathbb{N}_+)(\exists n \in \mathbb{N}_0)(\forall i < k) d_{n+i} = 9,$
CW	$\iff$	CW0 $\wedge$ CW9.

The decimal disjunctivity and normality nodes are

V1	$\iff$	$(\forall k \in \mathbb{N}_0)(\forall w \in \mathcal{D}_{10}^k)(\exists n \in \mathbb{N}_0)(\forall i < k) d_{n+i} = w_i,$
N10	$\iff$	$(\forall k \in \mathbb{N}_+)(\forall w \in \mathcal{D}_{10}^k) \lim_{N \rightarrow \infty} \frac{C_{10}(w, N; \pi)}{N} = 10^{-k}.$

The empty word is included in V1 to match the Lean definition; its condition is vacuous. Applying V1 to $0^M w$ shows that one occurrence of every word already implies infinitely many occurrences of each fixed word.

Absolute normality is

AN	$\iff$	$(\forall b \in \mathbb{N}_+)[b \geq 2 \Rightarrow (\forall k \in \mathbb{N}_+)(\forall w \in \mathcal{D}_b^k) \lim_{N \rightarrow \infty} \frac{C_b(w, N; \pi)}{N} = b^{-k}].$
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For a sequence $z = (z_n)_{n \geq 0} \subset [0, 1)$, let

$$\begin{aligned} \text{FA}(z) \iff & (\exists P \in \mathbb{N}_+)(\exists w_0, \dots, w_{P-1} \in [0, 1))(\forall \varepsilon > 0)(\exists K \in \mathbb{N}_0) \\ & (\exists t : \mathbb{N}_0 \rightarrow \{0, \dots, P-1\})(\forall k \in \mathbb{N}_0) \quad \|z_{K+k} - w_{t(k)}\|_{\mathbb{R}/\mathbb{Z}} < \varepsilon, \end{aligned}$$

and let UD(z) mean uniform distribution in $[0, 1)$. The Bailey–Crandall node is the global hypothesis

HA	$:\iff$	$(\forall b \in \mathbb{N}_+)(\forall p, q \in \mathbb{Z}[X])$ $[b \geq 2 \wedge p \neq 0 \wedge \deg p < \deg q \wedge (\forall n \in \mathbb{N}_+) q(n) \neq 0]$ $\Rightarrow [\text{FA}(z^{b,p,q}) \vee \text{UD}(z^{b,p,q})],$ $z_0^{b,p,q} = 0, \quad z_n^{b,p,q} = \left\{ bz_{n-1}^{b,p,q} + \frac{p(n)}{q(n)} \right\}.$
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Unlike every other node, **HA** is a global assertion about a class of recurrences, not a digit property of π .

Finally, the endpoint node is renamed to avoid collision with Theorem E of the companion article:

$$\boxed{\text{END} \iff \liminf_{n \rightarrow \infty} \|10^n \pi\|_{\mathbb{R}/\mathbb{Z}} = 0.}$$

Proposition 1.1 (Endpoint recurrence). [STATUS: PROOF SKETCH]

$$\text{CW0} \iff \liminf_{n \rightarrow \infty} x_n = 0, \quad \text{CW9} \iff \limsup_{n \rightarrow \infty} x_n = 1,$$

and

$$\boxed{\text{END} \iff (\text{CW0} \vee \text{CW9}).}$$

Proof. The two cylinder identities give the pointwise equivalences. Starts of arbitrarily long constant blocks must be unbounded, since a fixed all-zero or all-nine tail would contradict irrationality and the canonical expansion. Moreover

$$\|10^n \pi\|_{\mathbb{R}/\mathbb{Z}} = \min(x_n, 1 - x_n).$$

Thus either directed endpoint property implies **END**. Conversely, a sequence along which this minimum tends to zero has an infinite subsequence approaching either 0 or 1. $\square$

For every real α and every integer $b \geq 2$, define

$$\begin{aligned} \text{D}_b(\alpha) &\iff (\forall k \in \mathbb{N}_0)(\forall w \in \mathcal{D}_b^k)(\exists n \in \mathbb{N}_0)(\forall i < k) d_{n+i}^{(b)}(\alpha) = w_i, \\ \text{N}_b(\alpha) &\iff (\forall k \in \mathbb{N}_+)(\forall w \in \mathcal{D}_b^k) \lim_{N \rightarrow \infty} \frac{C_b(w, N; \alpha)}{N} = b^{-k}. \end{aligned}$$

The unparameterized nodes are the specializations at $\alpha = \pi$: for $b \in \{2, 16\}$, put $\text{D}_b := \text{D}_b(\pi)$ and $\text{N}_b := \text{N}_b(\pi)$. These abbreviations give the four nodes $\text{D}_2, \text{N}_2, \text{D}_{16}, \text{N}_{16}$.

Proposition 1.2 (Base-power equivalence). [STATUS: PROOF SKETCH] *For every real α , integer $b \geq 2$, and $k \geq 1$,*

$$\text{N}_b(\alpha) \iff \text{N}_{b^k}(\alpha), \quad \text{D}_b(\alpha) \iff \text{D}_{b^k}(\alpha).$$

In particular,

$$\text{N}_2 \iff \text{N}_{16}, \quad \text{D}_2 \iff \text{D}_{16}.$$

Proof. The normality assertion is Schmidt's theorem for multiplicatively dependent bases [23]. For disjunctivity, base- b^k digits are aligned blocks of k base- b digits. A base- b word can be padded to length divisible by k . Conversely, if u is the base- b unpacking of a base- b^k target, concatenate k copies of u , separated by one fixed digit. Their starting offsets run through every residue class modulo k , so one copy is aligned in every occurrence of the enlarged word. $\square$

1.3. Authoritative implication matrix. For numerical nodes, an arrow is a theorem about the corresponding property of every real, specialized to π . The **HA** row and column are type-different: a question mark means that no substantive theorem relating the global recurrence hypothesis to that node is asserted. The matrix freezes only the listed inter-node implications; it does not exclude auxiliary nodes or equivalent formulations.

$\Rightarrow$	C0	C9	C	V	N10	D2	N2	D16	N16	AN	HA	END
C0	=	$\times^{Z0}$	$\times^{Z0}$	$\times^{Z0}$	$\times^{Z0}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{Z0}$	?	✓
C9	$\times^{Z9}$	=	$\times^{Z9}$	$\times^{Z9}$	$\times^{Z9}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{Z9}$	?	✓
C	✓	✓	=	$\times^{Z09}$	$\times^{Z09}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{Z09}$	?	✓
V	✓	✓	✓	=	$\times^{DV}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{DV}$	?	✓
N10	✓	✓	✓	✓	=	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	?	✓
D2	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	=	$\times^{DB}$	✓	$\times^{DB}$	$\times^{DB}$	?	$\times^{X2}$
N2	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	✓	=	✓	✓	$\times^{X2}$	?	$\times^{X2}$
D16	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	✓	$\times^{DB}$	=	$\times^{DB}$	$\times^{DB}$	?	$\times^{X2}$
N16	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	$\times^{X2}$	✓	✓	✓	=	$\times^{X2}$	?	$\times^{X2}$
AN	✓	✓	✓	✓	✓	✓	✓	✓	✓	=	?	✓
HA	?	?	?	?	?	✓	✓	✓	✓	?	=	?
END	$\times^{Z9}$	$\times^{Z0}$	$\times^{Z0}$	$\times^{Z0}$	$\times^{Z0}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{X10}$	$\times^{Z0}$	?	=

TABLE 1. The implication matrix for the listed nodes. Here C0, C9, C, and V mean CW0, CW9, CW, and V1. A check mark is an unconditional implication; $\times^W$ cites a witness below.

All check marks follow from normality implying positive word frequency, Propositions 1.1 and 1.2, absolute normality, transitivity, and the Bailey–Crandall chain

$$HA \implies N_{16} \iff N_2 \implies D_{16} \iff D_2.$$

For this chain, irrationality excludes the finite-attractor branch of the BBP recurrence; the remaining steps are the conditional hypothesis, vanishing-tail transfer, the orbit criterion for normality, rational scaling, and dependent bases [2, 14, 23].

The witness labels have the following fixed meanings. The canonical decimals

$$Z0 = 0.(10^1)(10^2)(10^3)\dots, \quad Z9 = 0.(89^1)(89^2)(89^3)\dots$$

separate the two endpoint directions, while

$$Z09 = 0.(10^1 19^1)(10^2 19^2)\dots$$

has both constant-word families but omits digit 2. The witnesses DV and DB enumerate all decimal or binary words, respectively, with growing zero blocks inserted so that zero frequency tends to one; they are disjunctive but not normal. Finally, Schmidt’s deleted-digit construction supplies a decimal-normal $X10$ whose base-4 digits are only 0, 1, so it is not binary-disjunctive, and a binary-normal $X2$ whose decimal digits, after a carry-free translation by $1/9$, are only 1, 2. Hence $X2$ has no decimal endpoint recurrence [23].

The division of labour between the two documents is now fixed. This note is π -specific: it specifies the digit targets, proves exact reformulations, audits the BBP, Machin, Bailey–Crandall, and carry-certificate routes, and records the property-admission and machine-checked interfaces. The companion article [A] supplies the common symbolic, fractal, Diophantine, and game-theoretic framework and proves Theorems A–E and B’, all restated in the following unnumbered section. Neither document turns a route identity, a separate full-dimension statement, or the machine-checked interface into decimal disjunctivity or either directed constant-word endpoint theorem for π .

NOTATION AND RESULTS FROM THE COMPANION ARTICLE

The label [A] denotes M. Richter's companion article *Decimal word avoidance, badly approximable separators, and the BA-ALA intersection problem* (2026). This section records exactly the notation and article results used below; internal cross-references in the quoted statements are replaced only by the definitions on this page.

Coding, avoidance, and dimension. Decimal expansions are canonical: endpoints use the terminating expansion. For $x \in [0, 1)$, let $d_j(x)$ be its j -th decimal digit, and put $\text{code}_{10}(\omega) = \sum_{j \geq 1} \omega_j 10^{-j}$. For a nonempty word $w = w_1 \cdots w_m$, define

$$\begin{aligned}\Sigma_w &= \{\omega \in \mathcal{D}_{10}^{\mathbb{N}_+} : \omega_j \cdots \omega_{j+m-1} \neq w \ \forall j \geq 1\}, \\ K_w &= \text{code}_{10}(\Sigma_w), \\ C_w &= \{x \in [0, 1) : d_j(x) \cdots d_{j+m-1}(x) \neq w \ \forall j \geq 1\}.\end{aligned}$$

Put $\mathcal{D}_{10} = \{0, \dots, 9\}$. For a finite word P , let $\text{val}_{10}(P)$ be the integer represented by P , set $I_{10}(P) = [\text{val}_{10}(P)10^{-|P|}, (\text{val}_{10}(P) + 1)10^{-|P|})$, let $[P]_\Sigma$ be the symbolic prefix cylinder, and put $X_P = \text{code}_{10}(\Sigma_w \cap [P]_\Sigma)$. The prefix P is *admissible* when $\Sigma_w \cap [P]_\Sigma \neq \emptyset$; for this one-word shift, this is equivalent to internal w -avoidance. With $\mathcal{E}_{10} = \{j10^{-N} : N \in \mathbb{N}_0, 0 \leq j \leq 10^N\}$, the endpoint bridge is

$$C_w \triangle (K_w \cap [0, 1)) \subseteq \mathcal{E}_{10}, \quad (X_P \cap [0, 1)) \triangle (C_w \cap I_{10}(P)) \subseteq \mathcal{E}_{10}.$$

If ρ_w is the Perron–Frobenius eigenvalue of the prefix automaton for w -avoidance, then

$$d_w := \dim_{\mathbb{H}} C_w = \dim_{\mathbb{H}} K_w = \frac{\log \rho_w}{\log 10}, \quad 9 \leq \rho_w < 10.$$

Choose a safe guard digit and its reset word by

$$\beta \in \{1, \dots, 8\} \setminus \{w_1, w_m\}, \quad g = \beta^m.$$

Reading g after any w -free finite word is legal and returns the prefix automaton to state 0.

ALA, Diophantine classes, and regularity. A proper alphabet for w is $\mathcal{A} \subset \mathcal{D}_{10}$ with $|\mathcal{A}| = 9$ and $w \notin \mathcal{A}^m$. Put

$$J_k = \{n \in \mathbb{N}_0 : 10^k + 1 \leq n < 10^{k+1}\}.$$

For $c_* \in \mathcal{A}$, the property $\text{ALA}_{\mathcal{A}, c_*}$ is

$$\exists k_0 \ \forall k \geq k_0 \ \forall u \in \mathcal{A}^k \ \exists n \in J_k : \quad d_{n+1}(x) \cdots d_{n+k+1}(x) = uc_*.$$

The ledger notation $\text{ALA}_{\mathcal{A}, c}$ is the same definition with $(\mathcal{A}, c_*) = (\mathcal{A}, c)$. The interior-marker hypothesis is $c_* \in \mathcal{A} \cap \{1, \dots, 8\}$. Theorems B, C, E, and B' use it; Theorem D assumes only $c_* \in \mathcal{A}$. Write $\text{ALA}_P = X_P \cap \text{ALA}_{\mathcal{A}, c_*}$, and let $F_{j,P}$ be the compact projection of the sequences in $\Sigma_w \cap [P]_\Sigma$ satisfying the level- k ALA obligation for every $k \geq j$; then $\text{ALA}_P = \bigcup_{j \geq 1} F_{j,P}$. The weaker abundance set replaces uc_* by u , with the same windows J_k . For $\kappa > 0$,

$$\text{BA}(\kappa) = \left\{x \in \mathbb{R} : \left|x - \frac{p}{q}\right| \geq \frac{\kappa}{q^2} \ \forall p \in \mathbb{Z}, q \in \mathbb{N}_+\right\}, \quad \text{BA} = \bigcup_{M \geq 1} \text{BA}(1/M).$$

Write Trans for the transcendental reals. For irrational x ,

$$\mu(x) = \sup \left\{ \tau > 0 : 0 < \left|x - \frac{p}{q}\right| < q^{-\tau} \text{ for infinitely many } (p, q) \in \mathbb{Z} \times \mathbb{N}_+ \right\}.$$

Write $a_n(x)$ for its continued-fraction partial quotients, $p_n(x)/q_n(x)$ for its convergents, and $I_n^{\text{cf}}(x)$ for the continued-fraction cylinder determined by the first n partial quotients. A Borel measure ν is δ -Ahlfors regular if $cr^\delta \leq \nu(B(x, r)) \leq Cr^\delta$ for all support points x and all sufficiently small $r > 0$, for some $c, C > 0$. Define

$$\dim_{\mathbb{R}} E := \sup \left(\{0\} \cup \{\delta > 0 : \text{there is a } \delta\text{-Ahlfors-regular } \nu \text{ with } \text{supp } \nu \subset E\} \right).$$

For Theorem D, $K_{L,P}$ is [A]'s guarded regular subsystem, $s_L = \dim_{\mathbb{H}} K_{L,P}$, and $s_L \rightarrow d_w$.

Constant-run bound. Let $Z_x(n)$ and $N_x(n)$ be the lengths of the runs of zeros and nines, respectively, beginning immediately after the n -th decimal digit. If x is irrational and $\mu(x) \leq M < \infty$, then for every $\varepsilon > 0$,

$$Z_x(n), N_x(n) \leq (M - 1 + \varepsilon)n$$

for all sufficiently large n .

([A], Constant-run bound) If $x \in \text{BA}(\kappa)$, the fixed Diophantine constant sharpens this to $Z_x(n), N_x(n) \leq n + O_x(1)$.

Imported statements. In Theorems C and D, P is an arbitrary fixed admissible prefix; the marker hypotheses specified above are in force. **Theorem A (Badly approximable separator).** For every nonempty forbidden word w and every admissible prefix P ,

$$\dim_{\text{H}}(C_w \cap I_{10}(P) \cap \text{BA} \cap \text{Trans}) = \dim_{\text{H}}(C_w \cap I_{10}(P)) = d_w.$$

In particular, the avoider in every admissible decimal cylinder contains a full-dimensional set of badly approximable transcendental numbers, all of irrationality exponent 2.

([A], Theorem A)

Theorem B (Full-dimensional ALA with irrationality exponent 2). Assume the proper-alphabet and interior-marker hypotheses defined above. Then, for every admissible prefix P ,

$$\dim_{\text{H}}(C_w \cap I_{10}(P) \cap \text{ALA}_{\mathcal{A}, c_*} \cap \text{Trans} \cap \{x : \mu(x) = 2\}) = d_w.$$

Equivalently,

$$\dim_{\text{H}}(\text{ALA}_P \cap \text{Trans} \cap \{x : \mu(x) = 2\}) = d_w.$$

([A], Theorem B)

Theorem B' (Sharp logarithmic partial-quotient law). For every admissible prefix P ,

$$\dim_{\text{H}} \left\{ x \in \text{ALA}_P \cap \text{Trans} : \limsup_{n \rightarrow \infty} \frac{\log a_{n+1}(x)}{\log \log q_n(x)} = 1 \right\} = d_w.$$

Every point in the displayed set has irrationality exponent 2 and is not badly approximable.

([A], Theorem B')

Theorem C (Regularity obstruction). Let E be either ALA_P or the weaker abundance set defined above. There is no nonzero δ -Ahlfors-regular Borel measure μ with $\delta > 0$ and $\text{supp } \mu \subset E$. Consequently, $\dim_{\text{R}} E = 0$. The same conclusion holds for every subset of either all-label set.

([A], Theorem C)

Theorem D (Potential-winning obstruction). For every $0 \leq \gamma < d_w$, neither ALA_P nor any compact tail $F_{j,P}$ is γ -potential-winning in the complete metric space X_P . More precisely, for every $0 \leq \gamma < d_w$ there is an $L \geq 1$ such that the guarded set $K_{L,P}$ satisfies

$$K_{L,P} \subset X_P \setminus \text{ALA}_P, \quad \dim_{\text{R}} K_{L,P} = s_L > \gamma.$$

The same set $K_{L,P}$ is disjoint from every $F_{j,P}$.

([A], Theorem D)

Theorem E (Full-dimensional non-BA subsets of the all-label set). For every admissible prefix P , there are sets $F, L \subset \text{ALA}_P$ such that

$$F \text{ is } F_{\sigma}, \quad L = \liminf_{k \rightarrow \infty} U_k = \bigcup_{j \geq 1} \bigcap_{k \geq j} U_k,$$

where each $U_k \subset X_P$ is relatively open and combines a finite-coordinate abundance event with a finite Diophantine event, and

$$F \cap \text{BA} = L \cap \text{BA} = \emptyset, \quad \dim_{\text{H}} F = \dim_{\text{H}} L = d_w, \quad \dim_{\text{R}} F = \dim_{\text{R}} L = 0.$$

([A], Theorem E)

2. THE DECIMAL BBP SHADOW

For $x \in \mathbb{R}$, write

$$\{x\} = x - \lfloor x \rfloor, \quad \|x\|_{\mathbb{R}/\mathbb{Z}} := \min_{k \in \mathbb{Z}} |x - k|.$$

For $q \in \mathbb{Q}$, $\text{den}(q)$ is its positive denominator in lowest terms. If $D \geq 1$, then $\text{res}_D(u)$ denotes the least residue of the integer u in $\{0, \dots, D-1\}$.

Put

$$R(k) := \frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \quad (k \geq 0).$$

The Bailey–Borwein–Plouffe identity is

$$(2.1) \quad \pi = \sum_{k=0}^{\infty} \frac{R(k)}{16^k}.$$

[1] It is useful to record the elementary positive integral

$$(2.2) \quad R(k) = \int_0^1 x^{8k} g(x) dx, \quad g(x) := 4 - 2x^3 - x^4 - x^5 = (1-x)(x^2+2)(x^2+2x+2).$$

Thus $R(k) > 0$ and $R(k+1) < R(k)$.

For $n \geq 1$, set

$$(2.3) \quad B_n := \sum_{k=0}^{n-1} \frac{R(k)}{16^k} = \frac{P_n}{D_n}, \quad \gcd(P_n, D_n) = 1, \quad D_n > 0,$$

and define the canonical residue

$$(2.4) \quad r_n := \text{res}_{D_n} ((10^n - 16)P_n).$$

Then

$$(2.5) \quad \|(10^n - 16)B_n\|_{\mathbb{R}/\mathbb{Z}} = \frac{\min(r_n, D_n - r_n)}{D_n}.$$

The point of (2.4) is not computational difficulty at a fixed n ; it is the absence of a theorem placing the associated residue-plus-defect lift in the required shrinking one-sided windows for infinitely many n .

Proposition 2.1 (BBP decimal-shadow proposition). [STATUS: MACHINE-CHECKED]

Lean declarations:

Theory.PiDigits.T199BBPShadowPack.bbp10-defect-bounds-and-exact-shadow
and

Theory.PiDigits.T199BBPShadowPack.bbp10-theta-formula-strictAnti-tendsto.

The first declaration also gives the exact fractional-part shadow and decimal lift used in (2.18) and (2.19). For every $n \geq 2$, let

$$(2.6) \quad E_n := (10^n - 16)(\pi - B_n) > 0.$$

(The subscripted scalar E_n is a local defect, not the endpoint statement END.) Then

$$(2.7) \quad \left| \|(10^n - 16)\pi\|_{\mathbb{R}/\mathbb{Z}} - \|(10^n - 16)B_n\|_{\mathbb{R}/\mathbb{Z}} \right| < \left(\frac{5}{8} \right)^n.$$

Moreover, with

$$(2.8) \quad \Theta_n := (10^{n+1} - 16)B_{n+1} - 10(10^n - 16)B_n,$$

the sequence $(\Theta_n)_{n \geq 2}$ is strictly decreasing and

$$(2.9) \quad \Theta_n \downarrow 144\pi, \quad \Theta_n - 144\pi = 10E_n - E_{n+1}.$$

Proof. By (2.2), for $k \geq n \geq 1$,

$$0 < R(k) < \frac{4}{8k+1} \leq \frac{4}{8n+1}.$$

Hence

$$0 < \pi - B_n < \frac{4}{8n+1} \sum_{k=n}^{\infty} 16^{-k} = \frac{4 \cdot 16^{1-n}}{15(8n+1)}.$$

For $n \geq 2$, multiplication by $10^n - 16 < 10^n$ gives

$$(2.10) \quad 0 < E_n < \frac{64}{15(8n+1)} \left(\frac{5}{8}\right)^n < \left(\frac{5}{8}\right)^n.$$

Distance to the nearest integer is 1-Lipschitz, so (2.7) follows.

Since $B_{n+1} = B_n + 16^{-n}R(n)$,

$$(2.11) \quad \Theta_n = 144B_n + (10^{n+1} - 16)16^{-n}R(n).$$

The second term tends to zero and $B_n \rightarrow \pi$, proving $\Theta_n \rightarrow 144\pi$. Also

$$(2.12) \quad \begin{aligned} \Theta_{n+1} - \Theta_n = & - (10^{n+1} - 160)16^{-n}R(n) \\ & + (10^{n+2} - 16)16^{-n-1}R(n+1). \end{aligned}$$

For $n \geq 2$, $R(n+1) < R(n)$ and

$$\frac{10^{n+2} - 16}{16} < 10^{n+1} - 160 \iff 2544 < 6 \cdot 10^{n+1}.$$

Thus (2.12) is negative. Finally, substituting

$$B_j = \pi - \frac{E_j}{10^j - 16}$$

into (2.8) gives (2.9) identically. □

The same multiplier produces an autonomous affine decimal orbit. Set

$$(2.13) \quad v_n := \{(10^n - 16)\pi\}, \quad c := \{144\pi\}.$$

The elementary bounds $3.14 < \pi < 22/7$ give

$$452 < 144\pi < 453, \quad 50 < 16\pi < 51,$$

so

$$c = 144\pi - 452 \in (0, 1).$$

Then

$$(2.14) \quad v_{n+1} = \{10v_n + c\}.$$

The branch with carry 7 has the fixed point

$$(2.15) \quad a := \frac{7-c}{9} = 51 - 16\pi \in (0, 1),$$

because $10a + c = a + 7$. If $x_n := \{10^n\pi\}$, then $\{16\pi\} = 16\pi - 50 = 1 - a$, and therefore

$$(2.16) \quad v_n = \{x_n + a\}.$$

[STATUS: MACHINE-CHECKED] Lean declaration:

`Theory.PiDigits.T199BBPShadowPack.bbp10_affine_fixedPoint.`

Consequently,

$$(2.17) \quad \begin{aligned} \text{CW0} &\iff \forall \eta > 0 \exists n : a < v_n < a + \eta, \\ \text{CW9} &\iff \forall \eta > 0 \exists n : a - \eta < v_n < a. \end{aligned}$$

[STATUS: MACHINE-CHECKED] Lean declarations:

`Theory.PiDigits.T199BBPShadowPack.bbp10_rightApproach_iff_piCW0`

and `Theory.PiDigits.T199BBPShadowPack.bbp10_leftApproach_iff_piCW9`. This labels only the two equivalences; neither CW0 nor CW9 is proved.

For the reverse implications it is enough to take $\eta < \min(a, 1 - a)$. A point immediately to the right of a cannot arise from the wrapped branch of $\{x_n + a\}$, while a point immediately to the left cannot arise from the unwrapped branch. Irrationality excludes endpoint equality.

Euclidean division gives

$$(2.18) \quad v_n = \left\{ \frac{r_n}{D_n} + E_n \right\}, \quad \begin{aligned} v_n \in (a, a + \eta) &\iff \frac{r_n}{D_n} + E_n \in (a, a + \eta) + \mathbb{Z}, \\ v_n \in (a - \eta, a) &\iff \frac{r_n}{D_n} + E_n \in (a - \eta, a) + \mathbb{Z}. \end{aligned}$$

The bare residues are not equivalent to one-sided hits: the positive defect may cross a or an integer.

Unproved line BBP–10. One must show that these exact residue-plus-defect lifts enter arbitrarily small one-sided neighborhoods of $a = 51 - 16\pi$. Neither (2.7) nor monotonicity of Θ_n supplies that target-location theorem.

2.1. Oriented zero-block and all-cylinder conditions. Set

$$q_n := \frac{(10^n - 16)P_n - r_n}{D_n} \in \mathbb{Z}.$$

Then the definitions above give the exact lift

$$(2.19) \quad 10^n \pi = q_n + 51 + \left(\frac{r_n}{D_n} + E_n - a \right).$$

Problem 2.2 ($\text{SOH}_{\text{BBP},10}^0$). [STATUS: CONJECTURE]

$$\begin{aligned} (\forall k, N \in \mathbb{N}_+)(\exists n \in \mathbb{N}_0) \quad & n \geq \max\{N, 2\}, \\ & 0 < \frac{r_n}{D_n} + E_n - a < 10^{-k}. \end{aligned}$$

Equivalently,

$$a - E_n < \frac{r_n}{D_n} < a + 10^{-k} - E_n.$$

Put

$$t_n = \frac{r_n}{D_n} + E_n - a.$$

The BBP tail estimate gives, for $n \geq 2$,

$$0 < E_n < \eta_n, \quad \eta_n = (5/8)^n,$$

so $E_n \rightarrow 0$. Hence, for all sufficiently large n , $0 < E_n < a$, and therefore $-a < t_n < 1$. Given k, N , a CW0 hit may be chosen with $n \geq N$ sufficiently large and

$$0 < \{10^n \pi\} < \min\{10^{-k}, 1 - a\}.$$

The alternative lift $t_n = \{10^n \pi\} - 1$ would then give $t_n < -a$, which is impossible; hence $t_n = \{10^n \pi\}$. The reverse implication is immediate from (2.19). Thus

$$\boxed{\text{SOH}_{\text{BBP},10}^0 \iff \text{CW0.}}$$

[STATUS: MACHINE-CHECKED] Lean declaration:

`Theory.PiDigits.T199BBPShadowPack.bbp10_soh0_iff_piCW0`. This labels only the equivalence; neither $\text{SOH}_{\text{BBP},10}^0$ nor CW0 is proved.

This is an oriented condition, not a symmetric proximity statement.

The stronger all-cylinder condition is separate. Put

$$d_{\mathbb{T}}(x, y) = \|x - y\|_{\mathbb{R}/\mathbb{Z}}.$$

With the same η_n ,

$$d_{\mathbb{T}}\left(\{(10^n - 16)\pi\}, \frac{r_n}{D_n}\right) < \eta_n,$$

and, for an oriented arc J , define

$$J^{[-\eta]} = \{x \in J : d_{\mathbb{T}}(x, \partial J) > \eta\}, \quad J(v) = I_{10}(v) - 16\pi \pmod{1}.$$

Problem 2.3 (BBP-V1). [STATUS: CONJECTURE]

$$(\forall \ell \in \mathbb{N}_+)(\forall v \in \mathcal{D}_{10}^\ell)(\forall N \in \mathbb{N}_+)(\exists n \in \mathbb{N}_0) \\ n \geq \max\{N, 2\}, \quad 2\eta_n < 10^{-\ell}, \quad \frac{r_n}{D_n} \in J(v)^{[-\eta_n]}.$$

BBP-V1 implies arbitrary decimal-cylinder hitting and hence V1. It is a stronger route-specific sufficient condition, not another name for the oriented zero-block condition.

3. MACHIN 3/7 BRACKETS

Let

$$S_j(x) := \sum_{k=0}^j \frac{(-1)^k x^{2k+1}}{2k+1}.$$

For $m \geq 0$, define

$$(3.1) \quad \begin{aligned} L_m &:= 8S_{2m+1}(1/3) + 4S_{2m+1}(1/7), \\ U_m &:= 8S_{2m}(1/3) + 4S_{2m}(1/7). \end{aligned}$$

Proposition 3.1 (exact Machin brackets). [STATUS: MACHINE-CHECKED] *Lean declarations:*

Theory.PiDigits.HuttonRationalShadow.pi_eq_hutton
and *Theory.PiDigits.T198MachinBracketPack.machin37_strict_bracket_and_width*. One has

$$(3.2) \quad \pi = 8 \arctan(1/3) + 4 \arctan(1/7),$$

$$(3.3) \quad L_m < \pi < U_m,$$

and the exact width

$$(3.4) \quad W_m := U_m - L_m = \frac{8}{(4m+3)3^{4m+3}} + \frac{4}{(4m+3)7^{4m+3}}.$$

Proof. Put $\alpha = \arctan(1/3)$ and $\beta = \arctan(1/7)$. The double-angle formula gives

$$\tan(4\alpha) = \frac{24}{7}, \quad \tan(2\beta) = \frac{7}{24}.$$

Hence the denominator in the tangent addition formula vanishes. The branch is fixed explicitly by $\arctan x < x$:

$$0 < 4\alpha + 2\beta < \frac{4}{3} + \frac{2}{7} = \frac{34}{21} < \pi.$$

Therefore $4\alpha + 2\beta = \pi/2$, proving (3.2). For $0 < x < 1$, the alternating-series inequalities give

$$S_{2m+1}(x) < \arctan x < S_{2m}(x).$$

Applying them at $1/3$ and $1/7$ proves (3.3), and

$$S_{2m}(x) - S_{2m+1}(x) = \frac{x^{4m+3}}{4m+3}$$

gives (3.4). $\square$

State the bracket version of the constant-word target without hiding a carry. Say that MC0 holds if, for every $\ell \geq 1$, there exist $m, n \geq 0$ and $k \in \mathbb{Z}$ such that

$$(3.5) \quad \frac{k}{10^n} \leq L_m < U_m < \frac{k}{10^n} + 10^{-(n+\ell)}.$$

Proposition 3.2 (M — MC0 is exactly CW0). [STATUS: MACHINE-CHECKED] *Lean declaration: `Theory.PiDigits.T198MachinBracketPack.machinMC0_iff_piCW0`.*

$$(3.6) \quad \text{MC0} \iff \text{CW0}.$$

The endpoint convergence, zero-block/cylinder bridge, and bracket/cylinder equivalence used below are each *machine-checked*, respectively, as `Theory.PiDigits.T198MachinBracketPack.tendsto_machin37_endpoints`, `Theory.PiDigits.T198MachinBracketPack.piZeroBlock_iff_decimalCylinder`, and `Theory.PiDigits.T198MachinBracketPack.machinMC0_iff_piZeroCylinder`. This labels only the equivalence; neither MC0 nor CW0 is proved.

Proof. If (3.5) holds, then $L_m < \pi < U_m$ places π in the decimal cylinder

$$\left[\frac{k}{10^n}, \frac{k}{10^n} + 10^{-(n+\ell)} \right),$$

so the next ℓ digits are zero. Conversely, suppose CW0 holds and fix ℓ . Choose a zero block of length ℓ after position n , and put $k = \lfloor 10^n \pi \rfloor$. Irrationality gives positive distances from π to both endpoints of that cylinder. Since $L_m \uparrow \pi$ and $U_m \downarrow \pi$, all sufficiently large m satisfy (3.5). $\square$

The denominator has an exact 5-adic component. Write $L_m = A_m/Q_m$ in lowest terms and put $M = 4m + 3$.

Lemma 3.3 (the 5-adic denominator law). [STATUS: PROOF SKETCH]

$$(3.7) \quad v_5(Q_m) = \lfloor \log_5(4m + 3) \rfloor.$$

Proof. In $\mathbb{Q}_5$,

$$(3.8) \quad L_m = \sum_{\substack{1 \leq d \leq M \\ d \text{ odd}}} (-1)^{(d-1)/2} \frac{8 \cdot 3^{-d} + 4 \cdot 7^{-d}}{d}.$$

Let $e = \lfloor \log_5 M \rfloor$. Every summand has 5-adic valuation at least $-e$, so $v_5(L_m) \geq -e$. After multiplication by 5^e , only indices with $v_5(d) = e$ can survive modulo 5. Since $5^e \leq M < 5^{e+1}$ and d is odd, the only possibilities are

$$d = 5^e \quad \text{and} \quad d = 3 \cdot 5^e \quad \text{when} \quad 3 \cdot 5^e \leq M.$$

For $d = 5^e \equiv 1 \pmod{4}$, the surviving residue is

$$8 \cdot 3^{-1} + 4 \cdot 2^{-1} \equiv 3 \pmod{5}.$$

For $d = 3 \cdot 5^e \equiv 3 \pmod{4}$, including the alternating sign and division by 3, it is

$$-\frac{8 \cdot 3^{-3} + 4 \cdot 2^{-3}}{3} \equiv 1 \pmod{5}.$$

Thus $5^e L_m \equiv 3$ or $4 \pmod{5}$, never 0. Hence $v_5(L_m) = -e$, which is equivalent to (3.7). $\square$

Lemma 3.3 identifies exactly how much of the denominator a decimal power can kill automatically; it says nothing about the Archimedean size of the remaining residue. Let

$$\rho_{m,n} := \text{res}_{Q_m}(10^n A_m).$$

The following **small one-sided hit** condition is sufficient for an ℓ -zero certificate:

$$(\text{SOH})_{m,n,\ell} \quad \frac{\rho_{m,n}}{Q_m} + 10^n W_m < 10^{-\ell}.$$

Indeed,

$$\{10^n L_m\} = \frac{\rho_{m,n}}{Q_m}, \quad 0 < 10^n(\pi - L_m) < 10^n W_m,$$

so $(\text{SOH})_{m,n,\ell}$ leaves enough room for the whole scaled bracket $[10^n L_m, 10^n U_m]$ before the right endpoint of a length- ℓ zero cylinder. Thus SOH implies (3.5), and SOH for every ℓ implies CW0. SOH is a stronger sufficient condition based on the width bound; it is not identical to the exact defect event.

3.1. The quantified 3/7 condition. Let

$$D_m^M = \text{lcm}(\text{den } L_m, \text{den } U_m),$$

$$R_{m,n}^M = \text{res}_{D_m^M}(10^n D_m^M L_m), \quad \Delta_m = D_m^M(U_m - L_m).$$

Problem 3.4 $(\text{SOH}_{3/7})$. [STATUS: CONJECTURE]

$$(\forall k, M \in \mathbb{N}_+)(\exists m, n \in \mathbb{N}_0) \quad m \geq M \\ \wedge 10^k R_{m,n}^M + 10^{n+k} \Delta_m < D_m^M.$$

The inequality puts the whole scaled bracket inside $[0, 10^{-k}]$; conversely, an interior CW0 hit permits arbitrarily accurate brackets with arbitrarily large m . Thus

$$\boxed{\text{SOH}_{3/7} \iff \text{MC0} \iff \text{CW0}.$$

4. THE BAILEY–CRANDALL BASE-16 ORBIT

Let $R(n)$ be as in (2.1) and g as in (2.2), and define

$$(4.1) \quad \tau_n := \int_0^1 \frac{x^{8n}(1-x)(x^2+2)(x^2+2x+2)}{16-x^8} dx \quad (n \geq 0).$$

The integrand is positive on $(0, 1)$, so $\tau_n > 0$. Since $g(x) \leq 4$ and $16 - x^8 \geq 15$,

$$0 < \tau_n \leq \frac{4}{15(8n+1)},$$

and hence $\tau_n \rightarrow 0$.

Proposition 4.1 (Bailey–Crandall shadowing proposition). [STATUS: PROOF SKETCH]
 For every $n \geq 0$,

$$(4.2) \quad R(n) = 16\tau_n - \tau_{n+1}.$$

Put

$$(4.3) \quad Y_0 := 0, \quad Y_n := \sum_{k=0}^{n-1} 16^{n-1-k} R(k) \quad (n \geq 1),$$

and $y_n := \{Y_n\}$, so $y_0 = 0$. Then, for $n \geq 1$,

$$(4.4) \quad Y_n + \tau_n = 16^{n-1}\pi, \quad \{Y_n + \tau_n\} = \{16^{n-1}\pi\}.$$

Moreover,

$$(4.5) \quad y_{n+1} = \{16y_n + R(n)\} \quad (n \geq 0),$$

and the change of variables $z_n := \{y_n + \tau_n\}$ conjugates (4.5) to

$$(4.6) \quad z_{n+1} = \{16z_n\}.$$

Proof. Equation (4.2) follows directly from

$$16\tau_n - \tau_{n+1} = \int_0^1 x^{8n} g(x) \frac{16 - x^8}{16 - x^8} dx = R(n).$$

Using the geometric series under the positive integral,

$$\begin{aligned} \sum_{k=n}^{\infty} 16^{n-1-k} R(k) &= \int_0^1 g(x) \sum_{k=n}^{\infty} 16^{n-1-k} x^{8k} dx \\ &= \int_0^1 \frac{x^{8n} g(x)}{16 - x^8} dx = \tau_n. \end{aligned}$$

Multiplying (2.1) by 16^{n-1} and separating the first n terms gives (4.4). Equation (4.5) follows from $Y_{n+1} = 16Y_n + R(n)$, and (4.2) gives

$$z_{n+1} = \{16y_n + R(n) + \tau_{n+1}\} = \{16(y_n + \tau_n)\} = \{16z_n\}.$$

□

For completeness, the differencing operation used in proposed close-return arguments does not eliminate the phase. For two **real lifts before reduction modulo 1** of

$$X_{j+1} = 16X_j + f_j$$

with the same forcing, initial values separated by h are separated by $16^r h$ after r steps. Their r -step displacements therefore differ by

$$(4.7) \quad h \mapsto (16^r - 1)h.$$

This is an identity, not a mixing estimate. A close return controls a multiple of the unknown phase; no inverse estimate follows.

The recurrence is genuinely in the rational-function class of Bailey–Crandall Hypothesis A. Direct simplification gives

$$R(n) = \frac{120n^2 + 151n + 47}{(2n+1)(4n+3)(8n+1)(8n+5)},$$

and, in the indexing beginning at $n = 1$,

$$(4.8) \quad R(n-1) = \frac{120n^2 - 89n + 16}{(2n-1)(4n-1)(8n-7)(8n-3)}.$$

The numerator has degree 2, the denominator degree 4, and the denominator has no positive-integer zero: its roots are $1/2$, $1/4$, $7/8$, and $3/8$. Thus (4.5), with $y_0 = 0$, is an applicable instance of the class stated in Bailey–Crandall, §1 [2].

Equation (4.4) gives, in the circle metric,

$$(4.9) \quad \|y_n - \{16^{n-1}\pi\}\|_{\mathbb{R}/\mathbb{Z}} \leq \tau_n \longrightarrow 0.$$

Therefore (y_n) and $(\{16^{n-1}\pi\})$ have the same limit set **on** $\mathbb{R}/\mathbb{Z}$ and are uniformly distributed simultaneously; no claim is made about literal limit sets of representatives in $[0, 1)$, where wrapping at $0 = 1$ can intervene. These vanishing-perturbation statements are the content used from Lagarias, Theorem 3.1 and Lemma 3.2, and from Bailey–Crandall, Theorems 2.1–2.4 [14, 2].

The finite-limit-point alternative is unconditionally excluded. Lagarias, Theorem 3.3, and the corresponding finite-attractor results of Bailey–Crandall show that a base- b shift orbit has finitely many torus limit points only in the rational/eventually-periodic case [14, 2]. Since π is irrational, neither $(\{16^{n-1}\pi\})$ nor, by (4.9), (y_n) has a finite torus limit set. Consequently Hypothesis A implies

$$\text{HA} \implies \text{UD16}.$$

The exact perturbation relation yields

$$\text{UD16} \iff \text{N16},$$

and dependent-base normality gives

$$\text{N16} \iff \text{N2}.$$

Thus

$$(4.10) \quad \text{HA} \implies \text{UD16} \iff \text{N16} \iff \text{N2},$$

with $\text{N16} \implies \text{D16} \iff \text{D2}$ and $\text{N2} \implies \text{D2}$. Every conditional verb matters.

Unproved line BC–16. The equidistribution alternative of Hypothesis A is not proved for this orbit. Furthermore, UD16, N16, and N2 are hexadecimal/binary conclusions; no theorem used here converts them into decimal V1, CW0, or CW9. The exact coboundary (4.2) removes the positive forcing term and exposes, rather than solves, the phase $\{16^{n-1}\pi\}$.

4.1. The base-16 density classification. Write UD16 when the sequence $(y_n)_{n \geq 0}$ is uniformly distributed modulo 1. The tail-density statement is

Conjecture 4.2 (P5). [STATUS: CONJECTURE]

$$\boxed{(\forall N \in \mathbb{N}_0) \quad \overline{\{y_n : n \geq N\}} = [0, 1].}$$

Equivalently, for every rational $0 \leq u < v \leq 1$ and every $N \in \mathbb{N}_0$, some $n \geq N$ satisfies $u < y_n < v$.

Proposition 4.3 (P5 classification). [STATUS: PROOF SKETCH]

$$\boxed{\text{P5} \iff \text{D16} \iff \text{D2}.}$$

Proof. Proposition 4.1, in particular the circle shadowing in (4.9), makes P5 equivalent to base-16 disjunctivity. Proposition 1.2 then gives the second equivalence. $\square$

Thus P5 is a route-specific formulation of the hexadecimal/binary density branch, not an independent decimal target.

The BBP series supplies the recurrence, while Bailey–Crandall and Lagarias supply the orbit framework, vanishing-tail transfer, and finite/dense discussion [1, 2, 14]. The finite-limit-set branch is excluded by irrationality, but infinitely many limit points are weaker than density. The reopening condition is an unconditional proof of the displayed tail quantifier; conversely, one open interval missed by one tail disproves P5.

5. COMMON LOCATION-AND-CARRY OBSTRUCTION AND ROUTE AUDITS

The first two routes are instances of the following exact observation.

Observation 5.1 (residue–lift identity with arbitrary target). [STATUS: PROOF SKETCH] Let $x \in \mathbb{R}$, $M \in \mathbb{Z}_{>0}$, and let $q = P/D \in \mathbb{Q}$ be reduced with $D > 0$. Put

$$r = \text{res}_D(MP), \quad k = \frac{MP - r}{D} \in \mathbb{Z}, \quad \Delta = M(x - q).$$

Then

$$(5.1) \quad Mx = k + \frac{r}{D} + \Delta.$$

For every interval $I \subset [0, 1)$,

$$(5.2) \quad \{Mx\} \in I \iff \frac{r}{D} + \Delta \in \bigcup_{h \in \mathbb{Z}} (h + I).$$

If $0 < \Delta < 1$, then $0 < r/D + \Delta < 2$, so only carries $h = 0, 1$ can occur. In particular, for $0 < \eta < 1$,

$$(5.3) \quad \{Mx\} < \eta \iff \frac{r}{D} + \Delta \in [0, \eta) \cup [1, 1 + \eta).$$

Proof. Equation (5.1) is Euclidean division. Since k is an integer, taking fractional parts gives (5.2); the final assertion follows from $0 \leq r/D < 1$ and $0 < \Delta < 1$. $\square$

The integer k in (5.1) is exactly computable from the rational data; it is not an unknown. Under $0 < \Delta < 1$, the only additional integer is the carry

$$h = \left\lfloor \frac{r}{D} + \Delta \right\rfloor \in \{0, 1\}.$$

The missing theorem is an infinite quantitative location statement for the fractional parts of the residue-plus-defect lifts.

For BBP–10, use $M = 10^n - 16$, $q = B_n$, and $\Delta = E_n < 1$ for $n \geq 2$. Then (5.2) gives $v_n = \{r_n/D_n + E_n\}$, with target $I = (a, a + \eta)$ or $I = (a - \eta, a)$, where $a = 51 - 16\pi$ and $0 < \eta < \min(a, 1 - a)$. Thus BBP targets the interior point a , not 0.

For Machin–10, use $M = 10^n$, $q = L_m$, and $\Delta = 10^n(\pi - L_m)$, where $0 < \Delta < 10^n W_m$; the interval is $[0, 10^{-\ell})$. SOH is a stronger sufficient condition using this upper bound and leaving one-sided room before the right endpoint.

For Bailey–Crandall–16, (4.4) is $16^{n-1}\pi = Y_n + \tau_n$. Equivalently, take $M = 16^{n-1}$, $q = Y_n/16^{n-1}$, and $\Delta = \tau_n$ in (5.1). The positive defect is an exact coboundary, but the base-16 phase remains.

5.1. Close-pair and orientation audits. A bare close-pair pigeonhole argument does not supply SOH. Use the termwise common denominator

$$\tilde{Q}_m := 3^M 7^M \operatorname{lcm}\{d : 1 \leq d \leq M, d \text{ odd}\}, \quad M = 4m + 3,$$

and write $L_m = \tilde{A}_m / \tilde{Q}_m$. Once its 5-part has been cancelled by a sufficiently large decimal power, the remaining modulus is still divisible by 3^M , hence by 9. A close pair of orbit residues controls

$$(5.4) \quad (10^u - 10^v) \tilde{A}_m = 10^v (10^{u-v} - 1) \tilde{A}_m \pmod{\tilde{Q}_m},$$

not the one-sided representative of $10^v \tilde{A}_m$ itself. Recovering the latter by the proposed argument would require division by $10^{u-v} - 1$, but

$$(5.5) \quad 10^t \equiv 1 \pmod{9} \quad (t \geq 1),$$

so $10^t - 1$ is not a unit in that residue ring. Passing later to a reduced denominator does not recover information already lost: pigeonhole controls the multiplied difference (5.4), not either factor, and Archimedean smallness cannot be pulled backward through an uncontrolled multiplier. This invalidates the specific instruction “find a close pair, divide by $10^{u-v} - 1$, and recover a small starting residue.” It does not rule out a different pigeonhole argument supplied with additional arithmetic information.

Unproved line Machin–10. One must prove SOH, or an equivalent one-sided residue-location theorem, for arbitrarily large ℓ . The exact width (3.4), the equivalence (3.6), and the valuation law (3.7) do not place $\rho_{m,n}/Q_m$ near zero. That Archimedean location statement is the remaining decimal difficulty.

Pigeonhole close pairs yield a small residue of

$$10^i c (10^{j-i} - 1),$$

not an anchored hit by either orbit point. The route-compatible example with modulus $D = 9$ and $c = 1$ has $10^t c \equiv 1 \pmod{9}$ for all t , so differences vanish while a lower endpoint may be missed.

Likewise, use the involution

$$\iota_D(r) = [-r]_D,$$

not literal valuation preservation. It satisfies

$$\gcd(r, D) = \gcd(\iota_D(r), D),$$

or equivalently, with $\nu_p(0) = +\infty$,

$$(5.6) \quad \min\{\nu_p(r), \nu_p(D)\} = \min\{\nu_p(\iota_D(r)), \nu_p(D)\} \quad (p \mid D).$$

Negation commutes with multiplication modulo D but reverses the oriented Archimedean inequality.

5.2. Positive-tail carry killing and the Erdős comparison. Erdős’s Lambert-series argument succeeds because it supplies the missing carry-killing input. The abstract criterion is immediate but useful.

Lemma 5.2 (positive-tail carry killing). [STATUS: PROOF SKETCH] *Let $X = \sum_{j \geq 0} u_j$ with $u_j \geq 0$. If $X = A/B \in \mathbb{Q}$ in lowest terms and there exist indices $H_N < J_N$ and integers M_N such that*

$$(5.7) \quad B \mid M_N, \quad M_N \sum_{j \leq H_N} u_j \in \mathbb{Z}, \quad M_N u_j \in \mathbb{Z} \quad (H_N < j \leq J_N),$$

and

$$(5.8) \quad 0 < M_N \sum_{j > J_N} u_j < 1,$$

then a contradiction follows.

Proof. The rationality assumption and $B \mid M_N$ make $M_N X$ an integer. Conditions (5.7) make its head and manufactured block integers, whereas (5.8) puts the remaining positive tail strictly between zero and one. $\square$

For $\mathcal{L}_t$, Erdős used CRT congruences to manufacture the divisible block, an averaging/counting selection to choose the translate, and exponential decay to establish (5.8); see the theorem and congruences (2)–(6) on pp. 63–66 of [5]. The combination, rather than formal denominator clearing alone, is decisive.

A Ramanujan series for $1/\pi$ shows why a superficially analogous positive-tail argument does not transfer:

$$(5.9) \quad \frac{1}{\pi} = \sum_{n=0}^{\infty} \binom{2n}{n}^3 \frac{42n+5}{2^{12n+4}}.$$

This is Ramanujan’s §13, equation (30) [20]. Put $C_n := \binom{2n}{n}^3$, and let $s_2(n)$ be the binary digit sum. Legendre’s formula and $s_2(2n) = s_2(n)$ give

$$(5.10) \quad v_2 \binom{2n}{n} = (2n - s_2(2n)) - 2(n - s_2(n)) = s_2(n), \quad v_2(C_n) = 3s_2(n).$$

Since $42n+5$ is odd, the reduced dyadic denominator exponent is $\lambda_n := 12n + 4 - 3s_2(n)$. The identity $s_2(n+1) = s_2(n) + 1 - v_2(n+1)$ yields $\lambda_{n+1} - \lambda_n = 9 + 3v_2(n+1) > 0$; hence 2^{λ_N} clears every term through N . The algebraic half of (5.7) is available.

The scale comparison is nevertheless fatal for this criterion. Exactly, $12n + 4 = 3(4n) + 4$, whereas $3s_2(n) = O(\log n)$. With $N_{\text{fac}} = 4n$, the denominator ledger has linear scale $3N_{\text{fac}}$ but the valuation cancellation is only logarithmic. Moreover, $\binom{2n}{n} \sim 4^n / \sqrt{\pi n}$, so the n -th positive summand in (5.9) is $\asymp 2^{-6n} n^{-1/2}$. A multiplier clearing terms through N has size $2^{12N - O(\log N)}$; multiplied by the first omitted term, it has size $2^{6N - O(\log N)} N^{-1/2}$, which diverges. Extending the integral block moves both linear scales and does not reverse the inequality. Thus the valuation law supplies block integrality but not the positive small tail required by Lemma 5.2. This proves failure of this criterion, not impossibility of every other use of Ramanujan’s series.

The common unresolved step is now precise: **prove infinitely many one-sided target hits for the canonical residue-plus-defect lifts, with the controlled defect smaller than the available one-sided margin.** For a fixed carry h_j , the exact condition is $r_j/D_j \in h_j + I_j - \Delta_j$. A convenient stronger sufficient formulation, when the bare residue is required to lie on the same target side before the positive defect is added, is $0 < \Delta_j < \eta_j$ together with placement in the remaining interval of width $\eta_j - \Delta_j$. BBP supplies a moving modulus and a defect $< (5/8)^n$; Machin supplies exact brackets, their width, and the 5-part of the denominator; Bailey–Crandall supplies an exact positive coboundary and excludes one conjectural branch. None locates the least real representative in the required interval. That location theorem, not another truncation identity, is the unresolved content.

5.3. The Ramanujan reciprocal route. Erdős's Lambert-series argument supplies this mechanism, not P3 for π : CRT gives a weighted integral block and positivity controls the tail [5]. For Ramanujan's series, let

$$c_n = (6n+1) \binom{2n}{n}^3, \quad F(z) = \sum_{n \geq 0} c_n z^n, \quad F(1/256) = \frac{4}{\pi}$$

[20]. The formal reciprocal

$$F(z)^{-1} = \sum_{n \geq 0} \rho_n z^n$$

has integer coefficients determined by

$$(5.11) \quad \rho_0 = 1, \quad \rho_n = - \sum_{k=1}^n c_k \rho_{n-k}.$$

Because

$$|F(z) - 1| \leq \frac{4}{\pi} - 1 < 1 \quad (|z| \leq 1/256),$$

the reciprocal converges there, and

$$(5.12) \quad \frac{\pi}{4} = \sum_{n \geq 0} \rho_n 256^{-n} = 1 + \sum_{n \geq 1} (\rho_n 5^{8n}) (10^8)^{-n}.$$

Lemma 5.3 (Ramanujan reciprocal-coefficient valuation). [STATUS: PROOF SKETCH]
For every $n \geq 1$,

$$(5.13) \quad v_2(\rho_n) = 3s_2(n), \quad v_{10}(\rho_n 5^{8n}) = 3s_2(n) = O(\log n).$$

Proof. Put $H(z) = F(z) - 1$. The formal identity

$$F(z)^{-1} = \sum_{j \geq 0} (-H(z))^j$$

gives, for $n \geq 1$,

$$\rho_n = \sum_{j=1}^n (-1)^j \sum_{\substack{n_1 + \dots + n_j = n \\ n_i \geq 1}} c_{n_1} \cdots c_{n_j}.$$

Legendre's formula gives

$$v_2(c_m) = 3s_2(m).$$

Every summand therefore has valuation

$$3 \sum_i s_2(n_i) \geq 3s_2(n),$$

with equality exactly when the binary addition $n_1 + \dots + n_j = n$ is carry-free. If $s = s_2(n)$, the number of carry-free ordered compositions into j positive parts is $j! S(s, j)$, where $S(s, j)$ is a Stirling number of the second kind. After division by 2^{3s} and reduction modulo 2, the $j = 1$ contribution is odd, while every contribution with $j \geq 2$ has even multiplicity. The quotient is therefore odd, proving $v_2(\rho_n) = 3s_2(n)$. Finally $8n \geq 3s_2(n)$, so multiplication by 5^{8n} gives the decimal valuation in (5.13). $\square$

A direct absolute-value ramp based on a uniform coefficient-growth estimate would require

$$(5.14) \quad |\rho_n| \leq C\Lambda^n \quad (n \geq 0), \quad \Lambda < 256.$$

The asymptotic

$$c_n \asymp 64^n / \sqrt{n}$$

does not imply this bound for ρ_n . Assume $C > 0$, $0 < \Lambda < 256$, and (5.14). For a tail beginning after coefficient index N and a target bound $(10^8)^{-h}$, the direct absolute estimate requires

$$(5.15) \quad L > \frac{N \log(5^8 \Lambda) + h \log(10^8) + O(1)}{\log(256/\Lambda)}.$$

For the benchmark $\Lambda = 64$, which requires a separate proof, this is

$$(5.16) \quad L > 12.2877 \dots N + 13.2877 \dots h + O(1),$$

not $3N + O(\log N)$. The signed reciprocal coefficients remove the automatic positive-tail mechanism; they do not prove that selected blocks or cancellations can never meet P3.

6. PROPERTY ADMISSION AND SUPPORTING RESULTS

Sections 2–5 audit the π -specific analytic routes and stop at explicit missing location, carry, or density inputs rather than digit conclusions. This section changes the level of analysis: it asks which general arithmetic, dynamical, and Diophantine properties are compatible with avoiding a prescribed word, and which supporting tools can admit or exclude those properties. The imported avoidance, abundance, bad-approximation, and dimension notation used in this comparison is fixed in the preliminary summary of [A]’s notation and results.

Throughout, $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and

$$\widehat{\mu}(m) = \int_{\mathbb{T}} e^{-2\pi i m t} d\mu(t), \quad \widehat{f}(m) = \int_{\mathbb{T}} f(t) e^{-2\pi i m t} dt.$$

For arbitrary reals, radix expansions are greedy: at a radix endpoint we use the terminating expansion, not the eventually- $(b-1)$ expansion. Proposition R1 is imported from uniqueness-set theory. Lemmas R3 and R4 are proved below, with R4 identified as the finite-Fourier specialization of the Livšic–Veech coboundary criterion. They are supporting results, not article contributions.

6.1. Base- b orbits and aligned-block coding. Fix an integer $b \geq 2$. Let

$$T_b : \mathbb{T} \rightarrow \mathbb{T}, \quad T_b x = bx \pmod{1}.$$

For the representative $x \in [0, 1)$, set

$$a_n^{(b)}(x) = \lfloor bT_b^{n-1}x \rfloor \in \{0, \dots, b-1\}, \quad n \in \mathbb{N}_+.$$

Thus $d_n^{(10)}(\pi)$ is zero-indexed for π , with $n \in \mathbb{N}_0$, whereas $a_n^{(b)}(x)$ is the one-indexed general base- b notation. Then

$$(6.1) \quad x = \sum_{n \geq 1} a_n^{(b)}(x) b^{-n}, \quad T_b^n x = \sum_{j \geq 1} a_{n+j}^{(b)}(x) b^{-j}.$$

For a word $u = u_1 \cdots u_m \in \{0, \dots, b-1\}^m$, define the half-open cylinder

$$[u]_b = \left[\sum_{j=1}^m u_j b^{-j}, \sum_{j=1}^m u_j b^{-j} + b^{-m} \right).$$

With the chosen expansion, u occurs beginning at digit $n+1$ exactly when $T_b^n x \in [u]_b$.

Disjunctivity and orbit density. A real number is *disjunctive in base b* if every finite word over $\{0, \dots, b-1\}$ occurs in its base- b expansion. Lagarias calls this property *digit-dense* [14, Definition 2.1]. One has

$$(6.2) \quad x \text{ is base-}b \text{ disjunctive} \iff \{T_b^n x : n \geq 0\} \text{ is dense in } \mathbb{T},$$

the density assertion in [14, Theorem 2.1]. Indeed, a dense orbit meets the nonempty interior of every cylinder. Conversely, every nonempty open interval contains a sufficiently short base- b cylinder, so occurrence of every word makes the orbit dense.

Write

$$K_b(x) = \overline{\{T_b^n x : n \geq 0\}}$$

for the forward orbit closure.

6.1.1. *Aligned-block coding.* For a nonempty word w , let

$$C_w^{(b)} := \{x \in [0, 1) : \text{the greedy base-}b \text{ expansion of } x \text{ avoids } w \text{ at every position}\}.$$

Let $w = w_1 \cdots w_\ell$, and set

$$B = b^\ell, \quad \kappa(w) = \sum_{r=1}^{\ell} w_r b^{\ell-r}, \quad D_w = \{0, \dots, B-1\} \setminus \{\kappa(w)\},$$

and define the deleted-digit self-similar set

$$(6.3) \quad X_w = \left\{ \sum_{j \geq 1} e_j B^{-j} : e_j \in D_w \text{ for every } j \right\} \subset [0, 1].$$

It is the compact attractor of

$$(6.4) \quad \phi_e(t) = \frac{e+t}{B}, \quad e \in D_w.$$

We write the same symbol for its image in $\mathbb{T}$.

There is an essential one-alignment caveat. Membership in X_w excludes w only at the starts of the ℓ -digit blocks used in the displayed base- B coding. It does not exclude a copy crossing two such blocks or beginning in another congruence class modulo ℓ . At a radix endpoint, it asserts the existence of the displayed coding, not a statement about every alternative coding. Thus X_w is generally a strict superset of the globally w -avoiding set.

Aligned-block containment lemma. If $x \in C_w^{(b)}$, then

$$(6.5) \quad K_b(x) \subseteq X_w.$$

Proof. Fix $n \geq 0$ and regroup the suffix after digit n into blocks of length ℓ . For $j \geq 1$, put

$$e_j = \sum_{r=1}^{\ell} a_{n+(j-1)\ell+r}^{(b)}(x) b^{\ell-r}.$$

Each block begins at the position $n + (j-1)\ell + 1$. Global avoidance therefore gives $e_j \neq \kappa(w)$, so $e_j \in D_w$. Regrouping the absolutely convergent series (6.1) yields

$$T_b^n x = \sum_{j \geq 1} \sum_{r=1}^{\ell} a_{n+(j-1)\ell+r}^{(b)}(x) b^{-((j-1)\ell+r)} = \sum_{j \geq 1} e_j B^{-j} \in X_w.$$

The alignment is reset at the first digit of each suffix. This reset is precisely why global avoidance implies membership for every orbit point, despite X_w itself testing only one alignment. Since $X_w \subset \mathbb{T}$ is compact, it contains the orbit closure. $\square$

At level N , X_w is covered by $(B-1)^N$ intervals of length B^{-N} ; hence

$$(6.6) \quad \lambda(X_w) \leq \left(\frac{B-1}{B} \right)^N \rightarrow 0.$$

Thus X_w has Lebesgue measure zero.

6.2. Proposition R1: Rajchman measures on orbit closures. A finite Borel measure μ on $\mathbb{T}$ is *Rajchman* if

$$\widehat{\mu}(m) \rightarrow 0 \quad (|m| \rightarrow \infty).$$

A pseudofunction is a distribution whose Fourier coefficients tend to zero. Consequently, a finite measure is a pseudofunction exactly when it is Rajchman.

A set $E \subset \mathbb{T}$ is a *set of uniqueness* (a U -set) if every trigonometric series converging to zero on $\mathbb{T} \setminus E$ is the zero series. A closed U -set supports no nonzero pseudofunction; see Kechris–Louveau [11, Chapters II–III]. The class of U -sets is hereditary.

We use two forms of the self-similar uniqueness theorem.

Salem–Zygmund theorem (modern homogeneous null form). Let E be a homogeneous perfect set with common ratio $0 < \alpha < 1/2$. Suppose that, after the standard affine normalization of $\text{conv}(E)$ to $[0, 1]$, its defining maps are

$$S_j(t) = \alpha t + a_j \quad (1 \leq j \leq q),$$

with

$$0 = a_1 < a_2 < \cdots < a_q = 1 - \alpha, \quad a_{j+1} - a_j \geq \alpha, \quad q\alpha < 1.$$

Then E is a U -set if and only if $\theta = \alpha^{-1}$ is a Pisot number and

$$(6.7) \quad a_j \in \mathbb{Q}(\theta) \quad (1 \leq j \leq q).$$

Salem and Zygmund are the historical source [22]; the detailed normalized formulation used here is Kechris–Louveau [11, Theorem III.4.1].

Varjú–Yu theorem. A nonsingleton self-similar set is a U -set if and only if it has Lebesgue measure zero and its defining IFS can, after conjugation by a similarity, be written

$$t \mapsto r^{\ell_j} t + a_j,$$

where r^{-1} is Pisot, $\ell_j \in \mathbb{Z}_{>0}$, $\gcd_j \ell_j = 1$, and $a_j \in \mathbb{Q}(r)$ for all j [24, Theorem 1.4].

Proposition 6.1 (R1; Rajchman orbit-closure criterion). [STATUS: LITERATURE-CHECKED] *For every integer $b \geq 2$ and $x \in \mathbb{T}$, the orbit closure $K_b(x)$ supports a Rajchman probability measure if and only if x is disjunctive in base b .*

Proof. If x is disjunctive, (6.2) gives $K_b(x) = \mathbb{T}$. Normalized Haar measure is supported on $K_b(x)$ and has Fourier transform 1 at zero and 0 at every nonzero integer, so it is Rajchman.

Suppose that x is not disjunctive. Some nonempty word w is absent, and (6.5) gives $K_b(x) \subseteq X_w$. Put $B = b^{|w|}$. The maps in (6.4) have common contraction B^{-1} , rational translations e/B , and null attractor by (6.6). The integer $B \geq 2$ is a Pisot number.

If $B = 2$, then D_w has one digit and X_w is a singleton, hence a U -set. Let $B \geq 3$. Varjú–Yu applies directly with

$$r = B^{-1}, \quad \ell_e = 1, \quad e/B \in \mathbb{Q} = \mathbb{Q}(r),$$

and shows that X_w is a U -set.

The classical theorem gives the same conclusion. Let $p = \min D_w$ and $q = \max D_w$. The convex hull of X_w is

$$\left[\frac{p}{B-1}, \frac{q}{B-1} \right].$$

After its affine normalization to $[0, 1]$, the maps become

$$(6.8) \quad y \mapsto \frac{y}{B} + a_e, \quad a_e = \frac{(B-1)(e-p)}{B(q-p)} \in \mathbb{Q}.$$

The extreme translations are 0 and $1 - B^{-1}$, successive translations differ by at least B^{-1} , and there are $B-1$ maps, so $(B-1)/B < 1$. Thus (6.7) applies with $\alpha = B^{-1}$ and $\theta = B$.

By heredity, $K_b(x)$ is a U -set. A Rajchman probability measure supported on $K_b(x)$ would be a nonzero pseudofunction supported on a closed U -set, a contradiction. $\square$

Remark (Fourier dimension). Fourier dimension zero excludes positive-power Fourier decay, but does not by itself exclude a measure whose transform tends to zero arbitrarily slowly. The U -set conclusion is stronger for the present purpose: it excludes every nonzero Rajchman measure, irrespective of rate.

Remark (Lyons). Lyons constructed a Rajchman probability measure carried by the entire set of numbers non-normal in base 2 [17]. This does not conflict with R1, which concerns Rajchman measures supported on a single orbit closure. The topological support of Lyons’s measure need not be one nondisjunctive orbit closure. Nonnormality does not imply nondisjunctivity; a disjunctive nonnormal point has $K_b(x) = \mathbb{T}$.

6.3. Lemma R3: uniform pruning for prefix-local decimal rules. A *causal decimal rule* assigns to each rule-compatible finite history $u \in \{0, \dots, 9\}^{<\mathbb{N}}$ a set of allowed next digits

$$A(u) \subseteq \{0, \dots, 9\}.$$

A stream $d_1 d_2 \dots$ is rule-compatible when

$$d_{n+1} \in A(d_1 \dots d_n) \quad (n \geq 0).$$

We first construct digit streams and then apply the decimal coding map. The map is at most two-to-one: its only failures of injectivity are the countably many terminating/eventually-nine endpoint pairs. After the construction we discard

those endpoint codings; this countable deletion does not affect dimension and ensures that every retained stream is the greedy expansion of the represented real.

Lemma 6.2 (R3; uniform pruning lemma). [STATUS: PROOF SKETCH] *Let A be a causal decimal rule.*

- (1) *If $|A(u)| \geq 3$ for every compatible history u , then, for every nonempty word w , the set of rule-compatible w -avoiding reals contains a set of Hausdorff dimension*

$$(6.9) \quad \frac{\log 2}{\log 10}.$$

- (2) *If $|A(u)| \geq 2$ for every compatible history u , then, for every digit δ and every $\ell \geq 2$, the set of rule-compatible reals avoiding δ^ℓ contains a set of Hausdorff dimension*

$$(6.10) \quad \frac{\ell - 1}{\ell} \frac{\log 2}{\log 10}.$$

- (3) *In each case, the transcendental points of finite irrationality exponent inside the selected set have the same Hausdorff dimension as that selected set.*

The dimensions in (6.9)–(6.10) concern selected Moran subfamilies. They are lower bounds for the full rule-compatible avoidance set, not computations of its dimension.

Proof. For part 1, build a subtree recursively. Suppose u is a selected history and no copy of $w = w_1 \cdots w_\ell$ has yet occurred. At most one next digit can create a copy ending at the new position: this is possible only when the last $\ell - 1$ digits of u equal $w_1 \cdots w_{\ell-1}$, in which case the unique dangerous digit is w_ℓ . Delete that digit if present and retain any two digits from the remainder of $A(u)$. Since $|A(u)| \geq 3$, this is always possible. Every branch is compatible and avoids w , and every selected vertex has exactly two children.

For part 2, at each position divisible by ℓ , retain one allowed digit different from δ ; such a digit exists because $|A(u)| \geq 2$. At every other position retain any two allowed digits. Every interval of ℓ consecutive positions contains a multiple of ℓ , at which the digit is not δ ; hence no branch contains δ^ℓ . There are

$$(6.11) \quad 2^{q_n}, \quad q_n = n - \left\lfloor \frac{n}{\ell} \right\rfloor,$$

selected prefixes of length n .

The dimension argument is common to both constructions. Put $q_n = n$ in part 1 and use (6.11) in part 2. At level n , the selected set E is covered by 2^{q_n} decimal cylinders of diameter 10^{-n} . Therefore

$$(6.12) \quad \dim_H E \leq \liminf_{n \rightarrow \infty} \frac{q_n \log 2}{n \log 10}.$$

Give each retained pair of children equal conditional mass $1/2$, and give the unique retained child mass one at a nonbranching level. Each symbolic cylinder of length n has mass 2^{-q_n} . Every exceptional endpoint branch has mass

$$\lim_{n \rightarrow \infty} 2^{-q_n} = 0,$$

because $q_n \rightarrow \infty$; hence the countable union of endpoint branches has mass zero. If

$$s < \liminf_{n \rightarrow \infty} \frac{q_n \log 2}{n \log 10},$$

then $2^{-q_n} \leq 10^{-ns}$ for all sufficiently large n . An interval of length between $10^{-(n+1)}$ and 10^{-n} meets only a bounded number of level- n decimal cylinders, so the pushed-forward measure satisfies

$$\nu(I) \leq C_s |I|^s.$$

The mass distribution principle gives $\dim_{\mathbb{H}} E \geq s$. Letting s approach the liminf and combining with (6.12) proves

$$\dim_{\mathbb{H}} E = \liminf_{n \rightarrow \infty} \frac{q_n \log 2}{n \log 10}.$$

This is $\log 2 / \log 10$ in part 1 and $((\ell - 1) / \ell)(\log 2 / \log 10)$ in part 2. This is a direct mass-distribution argument for a nonstationary Moran-type construction; compare Moran for nested interval constructions [18]. Hutchinson's fixed-IFS theorem will be used separately below for the stationary deleted-digit set $\mathcal{C}_\delta$.

For part 3, the algebraic numbers are countable. The Liouville numbers have Hausdorff dimension zero by Jarník's Hausdorff-measure theorem [10]. Removing the union of those two zero-dimensional sets from E does not change $\dim_{\mathbb{H}} E > 0$. Every remaining point is transcendental and non-Liouville, hence has finite irrationality exponent. $\square$

Caution. Finite irrationality exponent does not mean exponent 2. The proof controls only finiteness after deletion of a zero-dimensional exceptional set.

6.4. Lemma R4: the primitive-ray criterion. Call $h \in \mathbb{Z} \setminus \{0\}$ *b-primitive* if $b \nmid h$. Every nonzero integer has a unique representation $m = b^k h$ with $k \geq 0$ and h *b-primitive*.

Lemma 6.3 (R4; primitive-ray criterion). [STATUS: PROOF SKETCH] *Let $b \geq 2$, and let f be a finite mean-zero trigonometric polynomial on $\mathbb{T}$. The following are equivalent.*

(a) *There is $g \in L^1(\mathbb{T})$ such that*

$$(6.13) \quad f = g - g \circ T_b.$$

(b) *There is a trigonometric polynomial g satisfying (6.13).*

(c) *For every b-primitive $h \neq 0$,*

$$(6.14) \quad \sum_{j \geq 0} \widehat{f}(b^j h) = 0.$$

The sum is finite because f has finite Fourier support.

Proof. For $g \in L^1(\mathbb{T})$, partitioning $[0, 1)$ into the b inverse branches of T_b gives

$$(6.15) \quad \widehat{g \circ T_b}(m) = \mathbf{1}_{b|m} \widehat{g}(m/b).$$

Indeed, after $y = bx - r$ on the branch indexed by $0 \leq r < b$, the relevant factor is

$$\frac{1}{b} \sum_{r=0}^{b-1} e^{-2\pi i m r / b},$$

which is one if $b \mid m$ and zero otherwise. Hence (6.13) implies, and is equivalent to,

$$(6.16) \quad \widehat{g}(m) - \mathbf{1}_{b|m} \widehat{g}(m/b) = \widehat{f}(m).$$

Assume (a). If h is primitive, (6.16) at $h, bh, \dots, b^k h$ telescopes to

$$(6.17) \quad \widehat{g}(b^k h) = \sum_{j=0}^k \widehat{f}(b^j h).$$

The left side tends to zero by the Riemann–Lebesgue lemma. The right side is eventually constant because f has finite Fourier support, and its eventual value is the sum in (6.14). Thus (c) holds.

Assume (c). Set $\widehat{g}(0) = 0$, and for the unique representation $m = b^k h \neq 0$ with h primitive define

$$(6.18) \quad \widehat{g}(b^k h) = \sum_{j=0}^k \widehat{f}(b^j h).$$

Only finitely many primitive rays meet the support of $\widehat{f}$. On each such ray, (c) makes the partial sum (6.18) vanish beyond its last active frequency. Thus $\widehat{g}$ has finite support and defines a trigonometric polynomial. Subtraction of consecutive partial sums proves (6.16); at $m = 0$, (6.16) holds because $\widehat{f}(0) = 0$. Hence (b) holds. The implication (b) $\Rightarrow$ (a) is immediate. $\square$

Attribution. This elementary lemma is the finite-Fourier form of the toral-endomorphism coboundary analysis in Veech [25, Section 5, especially (5.5) and Theorem 5.6]. Veech’s Theorem 1.2 and Corollary 1.4 give the corresponding smooth periodic-orbit criterion [25, Theorem 1.2 and Corollary 1.4], extending the Livšic paradigm [16]. The elementary finite-Fourier proof above is not a new coboundary theorem.

6.4.1. *Application to the T189 sectors.* The only abstract consequence needed is the contrapositive of Lemma 6.3:

If a finite mean-zero sector polynomial has a nonzero primitive-ray sum, then it is not a scalar L^1 -coboundary for T_b .

The conditional premise must remain explicit.

T189 sector-residue hypothesis. [STATUS: PROOF SKETCH] For each of T189’s nine nonzero child-character sectors r , the sector polynomial F_r is a finite mean-zero trigonometric polynomial, and there exists a 10-primitive frequency

$$h_r \in \mathbb{Z} \setminus 10\mathbb{Z}$$

such that

$$\sum_{j \geq 0} \widehat{F}_r(10^j h_r) \neq 0.$$

The top-band residue computation is intended to certify this premise.

Under this hypothesis, no sector polynomial F_r can be written

$$F_r = G_r - G_r \circ T_{10}, \quad G_r \in L^1(\mathbb{T}).$$

If such an L^1 transfer existed, Lemma 6.3 would first furnish a trigonometric-polynomial transfer $\widetilde{G}_r$. The resulting identity of trigonometric polynomials would hold pointwise and give

$$\sum_{n=0}^{N-1} F_r(T_{10}^n x) = \widetilde{G}_r(x) - \widetilde{G}_r(T_{10}^N x).$$

Thus no exact scalar endpoint potential eliminates any one of these sectors by telescoping. This conclusion does not upgrade the residue computation beyond proof-sketch status, and it does not exclude auxiliary-state or nontrivial matrix-cocycle extensions, approximate coboundaries, or cancellation in a separately checked linear combination of sectors.

6.5. Property admission and directed decimal

information. The finite-irrationality-exponent consequence used in the table below is the constant-run bound stated in the preliminary summary.

6.5.1. *A precise deleted-digit separator.* Let w be a nonempty decimal word, and choose a digit δ occurring in w . Define

$$(6.19) \quad \mathcal{C}_\delta = \left\{ \sum_{n \geq 1} d_n 10^{-n} : d_n \in \{0, \dots, 9\} \setminus \{\delta\} \right\}.$$

This is the attractor of the stationary IFS

$$\psi_d(t) = \frac{d+t}{10}, \quad d \in \{0, \dots, 9\} \setminus \{\delta\},$$

consisting of nine similarities of ratio $1/10$. It satisfies the open set condition, and Hutchinson's fixed-IFS theorem [9] gives

$$\dim_{\mathrm{H}} \mathcal{C}_\delta = \frac{\log 9}{\log 10}.$$

The IFS is irreducible. In dimension one, proper affine subspaces are points. If a finite set were invariant under every ψ_d , each injective ψ_d would permute it; a strict contraction can permute a finite set only when that set is the singleton consisting of its fixed point, which is impossible because the nine maps have distinct fixed points.

Let Bad denote the badly approximable reals. Kleinbock–Weiss's dimension theorem assumes an absolutely decaying Federer measure; their self-similar application supplies the required power-law measure for an irreducible similarity IFS satisfying the open set condition [12, Corollary 1.2 and Section 7.1]. Kristensen–Thorn–Velani assume an irreducible finite family of contracting similarities satisfying the open set condition [13, Corollary 10]. Fishman's winning theorem assumes a finite Borel absolutely friendly measure on a compact support, and his self-similar corollary supplies that setting for an irreducible IFS satisfying the open set condition [6, Theorem 1 and Corollary 3]. These results give

$$(6.20) \quad \dim_{\mathrm{H}}(\mathcal{C}_\delta \cap \mathrm{Bad}) = \dim_{\mathrm{H}} \mathcal{C}_\delta = \frac{\log 9}{\log 10}$$

[12, 13]; Fishman's formulation makes $\mathcal{C}_\delta \cap \mathrm{Bad}$ winning for Schmidt's game played on $\mathcal{C}_\delta$ [6]. Every point of this intersection is irrational and therefore has a unique decimal expansion. That expansion omits δ , hence avoids w . Removing the countable set of algebraic numbers preserves (6.20). Consequently, the w -avoiding set contains transcendental badly approximable points of full Hausdorff dimension relative to $\mathcal{C}_\delta$. Every irrational badly approximable number has irrationality exponent exactly 2.

The statement of Theorem A in the preliminary summary strengthens this easy deleted-digit example to every nonempty forbidden word and every admissible cylinder. The statement of Theorem C there gives the separate regularity obstruction for all-label abundance; neither result alone supplies a simultaneous BA–ALA point.

In the table below, to “share an exact identity” means that a *distinct* replacement real satisfies the same literal equality with the same fixed right-hand side. That is impossible when the identity uniquely fixes π . This logical point is not evidence that π contains w : π itself could, for all presently proved information, be a word avoider.

Property or exact input	Can a w -avoiding real share it?	What directed decimal information it yields
Irrationality	Yes. The transcendental badly approximable points in (6.20) are irrational and avoid w .	It excludes eventual periodicity. It forces no specified cylinder, digit, or word.
Transcendence	Yes. Deleting the countable algebraic subset from $\mathcal{C}_\delta \cap \text{Bad}$ leaves dimension $\log 9 / \log 10$. The transcendence of π is classical [15].	No target direction follows: a full-relative-dimensional separator has the same property while omitting w .
Finite irrationality exponent	Yes. The separator already contains badly approximable points, hence points of finite exponent 2. Lemma R3 supplies a weaker rule-compatible separator with merely finite exponent.	Only the constant-run bound stated in the preliminary summary applies. For π , it gives $Z_\pi(n), N_\pi(n) \leq (6.103205334138 + \varepsilon)n$ eventually; it does not force a nonconstant word.
Badly approximable / exact exponent 2	Yes. $\mathcal{C}_\delta \cap \text{Bad}$ is full-dimensional relative to $\mathcal{C}_\delta$, and winning on $\mathcal{C}_\delta$; its transcendental part has the same dimension. These two properties are not equivalent: badly approximable implies exponent 2, but exponent 2 is weaker. Neither property is known for π .	Bad approximation $ x - p/q \geq c_x q^{-2}$ sharpens the constant-run bound stated in the preliminary summary to $Z_x(n), N_x(n) \leq n + O_x(1)$. Exact exponent 2 alone gives $(1+\varepsilon)n$ eventually. Neither forces w .
Machin identity, $\pi = 8 \arctan(1/3) + 4 \arctan(1/7)$	No distinct replacement shares the literal identity; it follows directly from tangent addition. See also [3].	Proposition 3.1 gives the exact 3/7 brackets, and Proposition 3.2 proves $\text{MC0} \iff \text{CW0}$. What remains unproved is the small one-sided residue hit $(\text{SOH})_{m,n,\ell}$, not the target conversion itself.
BBP base-16 identity, $\pi = \sum_{k \geq 0} 16^{-k} R(k)$, $R(k) = \frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6}.$	No distinct replacement shares the literal equality [1].	It gives random-access hexadecimal digit extraction. Proposition 2.1 and Observation 5.1 give the exact decimal residue-plus-defect lift. What remains unproved is the one-sided target hit near $a = 51 - 16\pi$. Bailey–Crandall Hypothesis A yields only the conditional base-16/base-2 chain recorded in Section 4.
Ramanujan and Chudnovsky $1/\pi$ -series	No distinct replacement shares a fixed exact series value [20, 4].	They provide extremely rapid rigorous approximations, hence finite-prefix certification. Approximation rate alone supplies no prescribed-cylinder recurrence. Target-rotated sign claims remain <i>proof sketches</i> .
Euler/Wallis products, including $\frac{\pi}{2} = \prod_{n \geq 1} \frac{(2n)^2}{(2n-1)(2n+1)}$, and Euler's sine product	No distinct positive replacement shares the same fixed Wallis value or the same normalized sine-product zero set [3].	Truncations give convergent approximations or enclosures. No cited theorem converts them into recurrence to a chosen decimal cylinder; endpoint and sign translations remain <i>proof sketches</i> .
Even-zeta identities, $\zeta(2n) = \frac{(-1)^{n+1} B_{2n} (2\pi)^{2n}}{2(2n)!}$	No distinct positive replacement shares the equality with the same $\zeta(2n)$ and Bernoulli number: the positive $2n$ -th root recovers π [3].	They reconstruct or approximate π . They supply no one-sided, target-dependent decimal recurrence or prescribed-word theorem.

The separator (6.19)–(6.20) rules out any universal implication from any one of the first four properties alone to occurrence of an arbitrary prescribed word. Exact identities are different—a distinct replacement cannot share them—but uniqueness of the real value is not itself a digit theorem. The residue-and-defect reductions in Sections 2–5 are exact; the remaining unproved input is the required one-sided Archimedean target-location estimate.

7. FULL OPEN-PROBLEM LEDGER

This section uses the canonical coding, avoidance, all-label abundance, bad-approximation, and dimension notation fixed in the preliminary summary of [A]’s notation and results. In particular, C_w is the canonical decimal w -avoider, $d_w = \dim_H C_w$, and $\text{ALA}_{A,c}$ is the frozen all-label fresh-window property. The entries below preserve the full mathematical quantifiers while route-specific definitions are kept in Sections 2–4. The ledger counts CW0 and D2 as the underlying unresolved nodes; P3 and P5 occur only in their equivalence classifications and are not additional independent problems.

7.1. P1. The BA–ALA intersection problems. Put

$$X(w, A, c) = C_w \cap \text{BA} \cap \text{ALA}_{A,c},$$

and freeze the common parameter prefix as

$$(7.1) \quad \mathfrak{P}_1 := \left\{ (m, w, A, c) : \begin{array}{l} m \in \mathbb{N}_+, \ w \in \mathcal{D}_{10}^m, \ A \subseteq \mathcal{D}_{10}, \ c \in \mathcal{D}_{10}, \\ |A| = 9, \ w \notin A^m, \ c \in A \cap \{1, \dots, 8\} \end{array} \right\}.$$

Because $|A| = 9$, the condition $w \notin A^m$ is equivalent to saying that the unique digit omitted from A occurs in w , exactly as in the proper-alphabet convention stated in the preliminary summary. These parameter conditions also guarantee the safe reset used below and the interior-marker hypothesis used by the imported ALA theorems.

The three distinct propositions are

Problem 7.1 (P1-FD). [STATUS: CONJECTURE]

$$\forall (m, w, A, c) \in \mathfrak{P}_1, \quad \dim_H X(w, A, c) = d_w,$$

Problem 7.2 (P1-PD). [STATUS: CONJECTURE]

$$\forall (m, w, A, c) \in \mathfrak{P}_1, \quad \dim_H X(w, A, c) > 0,$$

Problem 7.3 (P1-NE). [STATUS: CONJECTURE]

$$\forall (m, w, A, c) \in \mathfrak{P}_1, \quad X(w, A, c) \neq \emptyset.$$

Choose a digit d occurring in w . Then

$$(\mathcal{D}_{10} \setminus \{d\})^{\mathbb{N}_+} \subseteq \Sigma_w.$$

Its image in K_w has Hausdorff dimension $\log 9 / \log 10$, and the countable endpoint bridge between $K_w \cap [0, 1)$ and C_w gives

$$d_w \geq \frac{\log 9}{\log 10} > 0.$$

$$(\text{P1} - \text{FD}) \implies (\text{P1} - \text{PD}) \implies (\text{P1} - \text{NE}).$$

The localized full-dimension strengthening is

Problem 7.4 (P1-FD-loc). [STATUS: CONJECTURE]

$$\begin{aligned} & (\forall (m, w, A, c) \in \mathfrak{P}_1) (\forall P \in \bigcup_{\ell \geq 1} \mathcal{D}_{10}^\ell) \\ & C_w \cap I_{10}(P) \neq \emptyset \implies \dim_H (X(w, A, c) \cap I_{10}(P)) = d_w. \end{aligned}$$

For this one-word avoidance shift, $C_w \cap I_{10}(P) \neq \emptyset$ is equivalent to internal avoidance by P , since such a prefix can be followed by a safe reset digit. The admissibility formulation is retained because it extends unchanged to the general SFT setting of P1'.

The packet-and-compactness route, including its finite-stage theorem and exact fixed-Markov-constant reopening condition, is audited in Subsection 7.2 below.

The statements of Theorem A, Theorem B, Theorem C, Theorem D, Theorem E, and Theorem B' in the preliminary summary give separate largeness, regularity, game, and metric statements. None supplies a simultaneous BA-ALA point, so none resolves P1-FD, P1-PD, or P1-NE. Theorem E constructs large subsets of ALA outside BA; it is an obstruction to largeness-only arguments, not a counterexample to P1.

7.2. Audit of the packet-and-compactness route. Fix $(m, w, A, c) \in \mathfrak{P}_1$, an admissible prefix P , and the safe guard $g = \beta^m$ from the preliminary summary. The two conclusions in the next proposition must be distinguished: one controls all rational approximations up to a denominator cutoff, while the other preserves an entire continued-fraction prefix.

Proposition 7.5 (Closeness-only square-root guarantees). [STATUS: PROOF SKETCH]
Let $x \in \text{BA}(\kappa)$, let $\delta > 0$, and suppose $|x - y| \leq \delta$.

(1) If $q \leq \sqrt{\kappa/(2\delta)}$, then for every $p \in \mathbb{Z}$,

$$\left| y - \frac{p}{q} \right| \geq \frac{\kappa}{2q^2}.$$

(2) If $\delta < \kappa/(4q_n(x)^2)$, then $y \in I_n^{\text{cf}}(x)$. Thus, when y is irrational, x and y have the same first n partial quotients.

Consequently, two numbers sharing their first N canonical decimal digits enjoy the first guarantee for

$$q \leq \sqrt{\frac{\kappa}{2}} 10^{N/2},$$

and the continued-fraction prefix is preserved whenever

$$q_n(x) < \frac{\sqrt{\kappa}}{2} 10^{N/2}.$$

Proof. The triangle inequality gives

$$\left| y - \frac{p}{q} \right| \geq \left| x - \frac{p}{q} \right| - |x - y| \geq \frac{\kappa}{q^2} - \delta,$$

which proves the first assertion. The endpoints of $I_n^{\text{cf}}(x)$ are, up to order,

$$\frac{p_n}{q_n}, \quad \frac{p_n + p_{n-1}}{q_n + q_{n-1}}.$$

Since $q_{n-1} \leq q_n$ and $x \in \text{BA}(\kappa)$,

$$\text{dist}(x, \partial I_n^{\text{cf}}(x)) \geq \min \left\{ \frac{\kappa}{q_n^2}, \frac{\kappa}{(q_n + q_{n-1})^2} \right\} \geq \frac{\kappa}{4q_n^2}.$$

The strict bound on δ keeps y in the same cylinder. Sharing N decimal digits gives $|x - y| \leq 10^{-N}$, which yields the stated consequences. $\square$

The order $\delta^{-1/2}$ is sharp up to constants for guarantees based only on closeness. Applying the $\text{BA}(\kappa)$ lower bound and the upper convergent estimate gives $a_{n+1} < \kappa^{-1}$; hence q_{n+1}/q_n is bounded in terms of κ . For the first $q_n \geq \delta^{-1/2}$,

$$q_n \ll_{\kappa} \delta^{-1/2}, \quad \left| x - \frac{p_n}{q_n} \right| < q_n^{-2} \leq \delta.$$

Taking $y = p_n/q_n$ destroys every positive bad-approximation bound at denominator q_n and puts y on a boundary of $I_n^{\text{cf}}(x)$. This is a limitation on uniform conclusions from $|x - y| \leq \delta$, not a claim that every perturbation fails; a specially chosen perturbation may remain badly approximable at all denominators. Likewise,

$$\left| x - \frac{p_n}{q_n} \right| + \delta < \frac{1}{2q_n^2}$$

only ensures that p_n/q_n is a convergent of y by Legendre's theorem [8, Theorem 1.1]; it does not preserve the whole prefix. Prefix preservation comes from the distance to the cylinder boundary.

A finite family of finite-coordinate obligations supported after $|P|$ is *feasible relative to P* if some admissible extension R of P fixes enough coordinates that every continuation in X_R satisfies them.

Theorem 7.6 (Finite-stage compatibility). [STATUS: PROOF SKETCH] *There is $k_0 = k_0(w, P)$ such that, for every finite $S \subset \{k \geq k_0\}$, one can find an admissible prefix Q extending P and ending in g for which every $x \in X_Q$ satisfies the all-label obligations at all levels $k \in S$, and*

$$\dim_{\text{H}}(X_Q \cap \text{BA} \cap \text{Trans}) = d_w.$$

The same conclusion holds for every finite family of future finite-coordinate obligations feasible relative to P . No assertion is made when P has already fixed an obligation incompatibly.

Proof. For each k , linearize a de Bruijn cycle of order $k+1$ over A . The resulting word D_k , of length $9^{k+1} + k$, contains every member of A^{k+1} , hence every uc with $u \in A^k$. Since words over A avoid w and g resets the automaton, gD_kg is legal in reset surroundings.

Place the first digit of its left guard at decimal position $10^k + 2$. Then the designated occurrences in D_k have frozen start indices

$$10^k + m + 1, \dots, 10^k + m + 9^{k+1}.$$

Moreover,

$$\frac{|gD_kg|}{|J_k|} = \frac{9^{k+1} + k + 2m}{9 \cdot 10^k - 1} = \left(\frac{9}{10}\right)^k + o(1).$$

Since $9^{k+1} + k + 2m + 1 < 9 \cdot 10^k$ for all sufficiently large k , every designated start lies in J_k , and the packet, whose last digit is at position $10^k + 9^{k+1} + k + 2m + 1$, ends before 10^{k+1} . Packets for distinct levels are therefore disjoint. An infinite family with one such packet at each large level occupies $O(9^K)$ of the first 10^K digits and therefore has zero asymptotic density. Increase k_0 so that the packets also lie beyond P and an initial reset guard.

For finite S , append g to P , fill unspecified positions by the safe digit β , insert gD_kg at the stated positions for $k \in S$, and finish with g ; for $S = \emptyset$, take $Q = Pg$. The resulting Q is admissible, ends in state 0, and fixes a valid occurrence of every

uc required at each $k \in S$. Hence every point of X_Q satisfies those obligations. The statement of Theorem A in the preliminary summary, together with the endpoint bridge recorded there, gives

$$\dim_{\mathbb{H}}(X_Q \cap \text{BA} \cap \text{Trans}) = d_w.$$

For a general feasible family, take a witnessing admissible extension R , append g after its last constrained coordinate, and apply the same argument to $Q = Rg$. $\square$

The theorem does not diagonalize through infinitely many stages with one fixed Markov constant. For $M \geq 1$, let

$$\text{BA}_M^{\text{cf}} := \{x \in [0, 1] \setminus \mathbb{Q} : a_n(x) \leq M \text{ for all } n \geq 1\}.$$

These sets are compact and

$$\begin{aligned} \text{BA} \cap [0, 1] &= \bigcup_{M \geq 1} \text{BA}_M^{\text{cf}}, \\ \text{BA}_M^{\text{cf}} &\subset \text{BA} \left(\frac{1}{M+2} \right), \\ \text{BA}(\kappa) \cap [0, 1] &\subset \text{BA}_{\lceil \kappa^{-1} \rceil}^{\text{cf}}. \end{aligned}$$

Indeed, if $q_n \leq q < q_{n+1}$, the best-approximation property gives $\text{dist}(qx, \mathbb{Z}) \geq \text{dist}(q_n x, \mathbb{Z})$, so bounded partial quotients imply $q \text{dist}(qx, \mathbb{Z}) > 1/(M+2)$. Conversely, applying $x \in \text{BA}(\kappa)$ to p_n/q_n and using the upper convergent estimate gives $a_{n+1} < \kappa^{-1}$. Thus a fixed M is quantitatively equivalent to a fixed positive lower bound for $\inf_q q \text{dist}(qx, \mathbb{Z})$.

Suppose $Q_1 \prec Q_2 \prec \dots$ are nested prefixes forcing successively more packet obligations. Each X_{Q_r} meets BA , but the argument supplies no M for which

$$X_{Q_r} \cap \text{BA}_M^{\text{cf}} \neq \emptyset \quad \text{for every } r.$$

Only after such an M had been fixed would these be nested compact sets to which compactness applies. The union $\text{BA} = \bigcup_M \text{BA}_M^{\text{cf}}$ is not closed. Indeed, if $x = [0; a_1, a_2, \dots]$ has unbounded partial quotients and

$$x_r = [0; a_1, \dots, a_r, 1, 1, \dots],$$

then $x_r \in \text{BA}$, $x_r \rightarrow x \notin \text{BA}$, and along a subsequence

$$\inf_{q \geq 1} q \text{dist}(qx_r, \mathbb{Z}) \leq \frac{1}{\max_{1 \leq j \leq r} a_j} \rightarrow 0.$$

Thus the available fixed Markov constant can collapse in the limit.

Nor is a fixed bound hidden in the dimension statement. Countable stability gives only

$$d_w = \dim_{\mathbb{H}}(X_Q \cap \text{BA}) = \sup_{M \geq 1} \dim_{\mathbb{H}}(X_Q \cap \text{BA}_M^{\text{cf}}).$$

For every $\varepsilon > 0$, some $M = M(Q, \varepsilon)$ captures dimension greater than $d_w - \varepsilon$, but the supremum need not be attained and no single M is uniform through all packet stages.

What the packet-and-compactness route lacks is therefore a fixed- κ extension lemma through the prescribed zero-density packets. Concretely, one would need some $\kappa > 0$ and nested prefixes Q_r , with Q_r forcing the first r prescribed packets, such that

$$X_{Q_r} \cap \text{BA}(\kappa) \neq \emptyset \quad \text{for every } r.$$

Theorem 7.6 allows the constant to depend on the finite prefix, while Proposition 7.5 protects it only through denominators of order $10^{N/2}$ after N decimal digits have been fixed. For the full-dimension form of P1, this route would need the corresponding dimension-uniform statement—given $s < d_w$, choose $\kappa(s) > 0$ before branching and retain dimension at least s —or a sparse-blackout theorem for a Schmidt-type game. A different proof of P1 need not take this form.

7.3. P1'. Sparse forced blocks in finite-type subshifts. For a sequence $\mathbf{a} = (a_j)_{j \geq 1} \in \mathcal{D}_b^{\mathbb{N}^+}$, write

$$\text{code}_b(\mathbf{a}) = \sum_{j \geq 1} a_j b^{-j}.$$

For $b \in \mathbb{N}$ with $b \geq 2$, let SFT_b be the set of one-sided SFTs in $\mathcal{D}_b^{\mathbb{N}^+}$, and define

$$\text{FBD}(b) = \left\{ ((N_j, \ell_j, v_j))_{j \geq 1} : \begin{array}{l} N_j \in \mathbb{N}_0, \ell_j \in \mathbb{N}_+, v_j \in \mathcal{D}_b^{\ell_j}, \\ N_{j+1} \geq N_j + \ell_j \ (\forall j \geq 1) \end{array} \right\}.$$

For $\Sigma \in \text{SFT}_b$ and $\mathcal{F} \in \text{FBD}(b)$, put

$$\Sigma_{\mathcal{F}} = \{x \in \Sigma : x_{N_j+1} \cdots x_{N_j+\ell_j} = v_j \ \forall j \geq 1\},$$

$$K_{\Sigma, \mathcal{F}} = \text{code}_b(\Sigma_{\mathcal{F}}), \quad F_{\mathcal{F}} = \bigcup_j \{N_j + 1, \dots, N_j + \ell_j\}.$$

Call $\mathcal{F}$ *sparse and entropy-neutral* for Σ if

$$\Sigma_{\mathcal{F}} \neq \emptyset, \quad \frac{|F_{\mathcal{F}} \cap \{1, \dots, N\}|}{N} \longrightarrow 0, \quad \dim_{\text{H}} K_{\Sigma, \mathcal{F}} = \dim_{\text{H}} \text{code}_b(\Sigma).$$

With $d_{\Sigma} = \dim_{\text{H}} \text{code}_b(\Sigma)$, the clean proposition is

Problem 7.7 (P1'). [STATUS: CONJECTURE]

$$\begin{aligned} & \forall b \in \mathbb{N}, \ b \geq 2, \ \forall \Sigma \in \text{SFT}_b \ \forall \mathcal{F} \in \text{FBD}(b) \ \forall s \in \mathbb{R}, \\ & (\Sigma \neq \emptyset \wedge \mathcal{F} \text{ is sparse and entropy-neutral for } \Sigma \wedge 0 < s < d_{\Sigma}) \\ & \implies \exists \kappa > 0 : \quad \dim_{\text{H}}(K_{\Sigma, \mathcal{F}} \cap \text{BA}(\kappa)) \geq s. \end{aligned}$$

The strict lower bound excludes the vacuous $s = 0$ case. Since

$$\text{BA} = \bigcup_{M \geq 1} \text{BA}(1/M),$$

countable stability gives

$$(7.2) \quad \dim_{\text{H}}(K_{\Sigma, \mathcal{F}} \cap \text{BA}) = d_{\Sigma} \iff (\forall s < d_{\Sigma})(\exists M \in \mathbb{N}_+) \dim_{\text{H}}(K_{\Sigma, \mathcal{F}} \cap \text{BA}(1/M)) \geq s.$$

Thus branch-dependent constants can count if full dimension of the BA union is proved; constants without fixed-level dimension control do not suffice.

The stationary literature covers supports satisfying its stated hypotheses: Federer/decay or friendliness in Kleinbock–Weiss, power-law/regularity assumptions in Kristensen–Thorn–Velani, and suitable fractal supports in Fishman’s Schmidt-game framework [12, 13, 7]. Self-similar or graph-directed examples require the relevant separation, irreducibility, decay, Federer, or power-law conditions. None of these theorems covers every nonstationary absolute-position forced corridor in P1'.

7.4. P2. Dissolved into route-specific conditions. The former P2 heading does not name one independent problem. The exact Machin bracket event is MC0; its equivalence with CW0 is Proposition 3.2 (*machine-checked* as `Theory.PiDigits.T198MachinBracketPack.machinMC0_iff_piCW0`); the full quantified $\text{SOH}_{3/7}$ condition is Problem 3.4. The oriented decimal BBP condition is Problem 2.2, and the stronger all-cylinder condition is Problem 2.3. Thus

$$\text{SOH}_{3/7} \iff \text{MC0} \iff \text{CW0}, \quad \text{SOH}_{\text{BBP},10}^0 \iff \text{CW0},$$

whereas BBP-V1 implies V1 . The reopening conditions are the displayed oriented universal inequalities themselves. The close-pair, valuation, and orientation failures that prevent the current arithmetic inputs from proving them are recorded in Section 5.

7.5. CW0. The P3 carry-certificate classification. The unresolved ledger node here is CW0. P3 is the following named equivalent formulation, not an independent open problem.

Conjecture 7.8 (P3). [STATUS: CONJECTURE]

$$\begin{aligned} & \exists q, s, B \in \mathbb{N}_+ \exists Z \in \mathbb{Z} \exists (a_m)_{m \geq 1} \in \mathbb{Z}^{\mathbb{N}_+} \\ & \quad \exists (n_j, L_j, h_j)_{j \geq 1} \in (\mathbb{N}_+^3)^{\mathbb{N}_+} : \\ & \quad B = 10^s, \quad \frac{\pi}{q} = Z + \sum_{m \geq 1} a_m B^{-m}, \quad \sum_{m \geq 1} |a_m| B^{-m} < \infty, \\ & \quad n_1 < n_2 < \dots, \quad h_j \rightarrow \infty, \\ & \quad (\forall j \in \mathbb{N}_+) \quad B^{L_j} \mid \sum_{r=1}^{L_j} a_{n_j+r} B^{L_j-r} \\ & \quad \quad \quad \wedge 0 < \sum_{r > L_j} a_{n_j+r} B^{-r} < B^{-h_j}. \end{aligned}$$

Proposition 7.9 (P3 classification). [STATUS: PROOF SKETCH]

$$\boxed{\text{P3} \iff \text{CW0}.}$$

Proof. Assume P3. After multiplication by B^{n_j} , the weighted block is integral and

$$0 < \left\{ B^{n_j} \frac{\pi}{q} \right\} < B^{-h_j}.$$

For large j ,

$$0 < \{B^{n_j} \pi\} < qB^{-h_j},$$

giving at least

$$h_j - \lceil \log_B q \rceil$$

leading base- B zero digits. Since $B = 10^s$, P3 implies CW0.

Conversely, assume CW0. Take $q = 1$, $s = 1$, $B = 10$, $Z = 3$, and let $a_m \in \{0, \dots, 9\}$ be the canonical decimal digits of π . Choose increasing positions n_j at which zero runs of lengths $H_j \rightarrow \infty$ begin; such positions can be chosen increasing because π is not eventually zero. Put $L_j = 1$ and $h_j = H_j$. Then

$$10^{L_j} \mid a_{n_j+1} = 0,$$

and

$$0 < \sum_{r > 1} a_{n_j+r} 10^{-r} < 10^{-H_j}.$$

Positivity follows because π is irrational and hence not terminating. Strictness of the upper bound follows because the canonical expansion is not eventually 9. Therefore P3 holds. $\square$

The arithmetic value of this equivalence would come from coefficients and blocks derived independently of the target decimal-cylinder assertion. Section 5 records the positive-tail mechanism, the Ramanujan audit, and the precise sign, divisibility, and growth obstructions; in particular, Lemma 5.3 gives the exact valuation input and the discussion following it states the reopening condition.

7.6. P4. The eight constant-word components. Let $a_n^{(10)}(\alpha)$ be the n -th digit after the radix point in the canonical decimal expansion. Define

$$\text{CW}_\delta(\alpha) : (\forall k \in \mathbb{N}_+)(\exists n \in \mathbb{N}_0)(\forall i \in \mathbb{N}_+)(i \leq k \Rightarrow a_{n+i}^{(10)}(\alpha) = \delta).$$

Then

Problem 7.10 (P4). [STATUS: CONJECTURE]

$$\bigwedge_{\alpha \in \{\pi, \sqrt{2}, e, \log 2\}} \bigwedge_{\delta \in \{0, 9\}} \text{CW}_\delta(\alpha).$$

A proof of one component resolves only that component; a disproof of one component disproves P4. For irrational α ,

$$(7.3) \quad \begin{aligned} \text{CW}_0(\alpha) &\iff \liminf_n \{10^n \alpha\} = 0, \\ \text{CW}_9(\alpha) &\iff \limsup_n \{10^n \alpha\} = 1. \end{aligned}$$

Write

$$\text{CW0} = \text{CW}_0(\pi), \quad \text{CW9} = \text{CW}_9(\pi), \quad \text{CW} = \text{CW0} \wedge \text{CW9}.$$

For

$$\text{END} : \liminf_n \|10^n \pi\|_{\mathbb{R}/\mathbb{Z}} = 0,$$

the exact relation is

$$\text{END} \iff (\text{CW0} \vee \text{CW9}),$$

not CW.

Applying the constant-run bound stated in the preliminary summary to the conservative certified bound

$$M_\pi := 7.103205334138, \quad \mu(\pi) \leq M_\pi,$$

if $\ell_\delta^\pi(n)$ is the run length beginning after position n , then for every $\varepsilon > 0$,

$$(7.4) \quad \begin{aligned} \max\{\ell_0^\pi(n), \ell_9^\pi(n)\} &\leq (M_\pi - 1 + \varepsilon)n \\ &= (6.103205334138 + \varepsilon)n \quad \text{for all sufficiently large } n \end{aligned}$$

[26]. This bounds an existing run; it does not prove one exists. The statement of Theorem A in the preliminary summary gives full relative dimension d_{0^k} in C_{0^k} , and d_{9^k} in C_{9^k} , for transcendental badly approximable numbers. Since every badly approximable irrational has irrationality exponent 2, these sets also lie in $\{x : \mu(x) = 2\}$; they are not dimension-one claims.

For each of the eight conjuncts, the reopening condition is its displayed existence quantifier for arbitrarily large constant blocks. A proof resolves that component;

Item	Exact Lean namespace or definition
T191	Theory.PiDigits.T191CentralBoundaryKernelFloor
T192	Theory.PiDigits.T192PrimitiveValuationShells
T193	Theory.PiDigits.T193PositiveValuationShellAggregate
T194	Theory.PiDigits.T194CentralPiReturnSeed
T198	Theory.PiDigits.T198MachinBracketPack
T199	Theory.PiDigits.T199BBPShadowPack
V1	Theory.PiDigits.piDigit and Theory.PiDigits.V1

TABLE 2. Inventory of the Lean interface described below.

one proved missing block length disproves it. For the two π -components, Proposition 1.1 shows that END supplies only their disjunction and does not select a direction.

7.7. D2. The P5 base- 16 density classification. The unresolved ledger node here is D2, equivalently D16. Conjecture 4.2 gives the full tail quantifier, and Proposition 4.3 proves its equivalence with these underlying nodes. The finite-limit-set alternative is excluded for the π seed, but an infinite limit set need not be dense. Reopening therefore requires an unconditional density theorem for this specific orbit; a tail missing one nonempty open interval is a complete disproof. P5 is an equivalent hexadecimal/binary formulation, not an independent fifth decimal problem.

APPENDIX A. MACHINE-CHECKED INTERFACE

Notation and mathematical trust boundary. This appendix describes the Lean interface in <https://github.com/thepok/PI> at the commit pinned in Appendix B. The exact declaration names, hypotheses, conclusions, and axiom allowlist below delimit the machine-checked claim.

$$q_k = 10^k, \quad \xi_n = \text{fract}(10^n \pi), \quad c(q, A) = \text{decimalCylinderCenter } q \ A,$$

and abbreviate

$$S(q, A, n) = q \ \Re(\text{primitiveBoundaryFourierBlockSum } q \ A \ n \ 1) \\ - \text{signedBlockPotential } q.$$

Write $D(q, r, t) = \text{divisibleBoundaryPolynomial } q \ r \ t$ and $M(q, z) = \Re(\text{boundaryMinorant } q \ z)$. An identifier below consists of the displayed exact namespace prefix followed by the displayed exact suffix.

The central audit file registers `#print axioms` queries; the allowlist itself is enforced by `workflows/verification/check.ps1`. For every declaration marked $[A]$, the scripted audit rejects every dependency outside

$$\mathcal{A} = \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}.$$

Thus the dependency set is a subset of $\mathcal{A}$, not necessarily all three axioms. The dependency set recorded for each audited declaration is therefore a subset of this allowlist. The marker $[A]$ below records only this trust boundary; it is separate from, and is not, a claim-status label.

External conditional premise. [STATUS: MACHINE-CHECKED] The exact Lean name `Theory.PiDigits.LongLagBlockCollisionDecay.T4.IrrationalityMeasureBelow` defines

$$\begin{aligned} \text{IM}_b(x) &: \iff \exists \mu \in \mathbb{R}, \mu < b \wedge \\ &\quad \forall \varepsilon \in \mathbb{R}, 0 < \varepsilon \rightarrow \\ &\quad \exists Q_0 \in \mathbb{N}, \forall q \in \mathbb{N}, Q_0 \leq q \rightarrow 0 < q \rightarrow \\ &\quad \forall p \in \mathbb{Z}, \frac{1}{q^{\mu+\varepsilon}} < \left| x - \frac{p}{q} \right|. \end{aligned}$$

The order of these quantifiers is part of the definition: Q_0 may depend on ε , after one witness $\mu < b$ has been fixed. Below, H_8 and H_9 mean respectively $\text{IM}_8(\pi)$ and $\text{IM}_9(\pi)$.
T191: central kernel floor. Let $\mathbf{N}_{191}$ =
`Theory.PiDigits.T191CentralBoundaryKernelFloor`. [STATUS: MACHINE-CHECKED]
The theorem `N191.boundaryMinorant_re_gt.4859_div_10000` says: for every $k \in \mathbb{N}$ with $3 \leq k$, and every $y \in \mathbb{R}$ with $|y| \leq 9/22$,

$$M(q_k, y/q_k) > 4859/10000.$$

It has no irrationality-measure premise. [A]

T192: exact decimal-valuation shells. Let $\mathbf{N}_{192}$ =
`Theory.PiDigits.T192PrimitiveValuationShells`. The suffix `At` denotes an arbitrary real phase x ; the version without `At` substitutes $x = \xi_n$.

[STATUS: MACHINE-CHECKED] `N192.primitiveBoundaryFourierSum_succ_sub_eq_atom`: for all $q, A, n \in \mathbb{N}$, the primitive prefix sum at $n+1$ minus that at n equals `primitiveBoundaryAtom q A n`. [A]

[STATUS: MACHINE-CHECKED] `N192.primitiveBoundaryAtom_eq_sum_valuationShells_generic` and `N192.primitiveBoundaryAtom_eq_sum_valuationShells`: for all k, A and respectively all phases x , or all orbit times n , the atom at scale q_k is exactly the sum of shells with valuations $s = 0, \dots, k$. There is no lower-bound hypothesis on k . [A]

[STATUS: MACHINE-CHECKED] `N192.primitiveValuationShell_eq_layerDifference_generic` and `N192.primitiveValuationShell_eq_layerDifference`: for every q, A, s , and every x or n , the exact shell is

$$D(q, 10^s, x - 10^s c(q, A)) - D(q, 10^{s+1}, 10x - 10^{s+1} c(q, A)),$$

with $x = \xi_n$ in the orbit version. [A]

[STATUS: MACHINE-CHECKED] `N192.valuationZeroLayerDifference_re_ge`: if $q \geq 1000$ and $|z| \leq 9/(22q)$, then

$$\frac{9}{20} M(q, z) \leq \Re(D(q, 1, z) - D(q, 10, 10z)).$$

[A]

[STATUS: MACHINE-CHECKED] `N192.primitiveValuationShell_zero_re_ge_generic` and `N192.primitiveValuationShell_zero_re_ge`: if $k \geq 3$, $|y| \leq 9/22$, and the exact central-coordinate equation

$$x - c(q_k, A) = y/q_k \quad (\text{respectively, } \xi_n - c(q_k, A) = y/q_k)$$

holds, then the real part of the zero shell is at least $\frac{9}{20} M(q_k, y/q_k)$. The generic theorem is centrally audited. The orbit theorem is exported and locally queried by `#print axioms` in T192, but is omitted from `audit/AxiomAudit.lean` at the cited commit; no central-audit claim is made for that one.

[STATUS: MACHINE-CHECKED] `N192.primitiveValuationShell_zero_re_gt_generic` and `N192.primitiveValuationShell_zero_re_gt`: under the same three hypotheses, the zero-shell real part is strictly greater than $\frac{9}{20} \frac{4859}{10000}$. [A] No theorem in T192 takes H_8 , H_9 , or `irrational.pi` as a premise.

T193: positive-shell aggregate and deterministic central seed. Let $N_{193} = \text{Theory.PiDigits.T193PositiveValuationShellAggregate}$.

[STATUS: MACHINE-CHECKED] $N_{193}.\text{positiveValuationLayerMassSum.lt.five.div.eighteen}$: for $k \geq 3$, the sum of $\text{boundaryLayerMass } q_k 10^s$ over $1 \leq s \leq k$ is strictly below $5/18$. [A]

[STATUS: MACHINE-CHECKED] $N_{193}.\text{positiveValuationShellSum.re.gt.generic}$ and $N_{193}.\text{positiveValuationShellSum.re.gt}$: for $k \geq 3$, every A , and every phase x or orbit time n , the real part of the sum of shells $1 \leq s \leq k$ is greater than $-108019/1800000$. No central-coordinate hypothesis is needed. [A]

[STATUS: MACHINE-CHECKED] $N_{193}.\text{primitiveBoundaryAtom.re.gt.7139.div.45000.generic}$ and $N_{193}.\text{primitiveBoundaryAtom.re.gt.7139.div.45000}$: if $k \geq 3$, $|y| \leq 9/22$, and $x - c(q_k, A) = y/q_k$ (respectively $\xi_n - c(q_k, A) = y/q_k$), then the atom's real part is greater than $7139/45000$. [A]

[STATUS: MACHINE-CHECKED] $N_{193}.\text{central.unitBlock.surplus.gt.three.div.twenty.generic}$ and $N_{193}.\text{central.unitBlock.surplus.gt.three.div.twenty}$: under exactly the same hypotheses,

$$q_k \Re(\text{primitiveBoundaryAtomAt } q_k A x) - \text{signedBlockPotential } q_k > \frac{3}{20} q_k,$$

or, in the orbit version, $\mathcal{S}(q_k, A, n) > \frac{3}{20} q_k$. [A] No T193 theorem uses an irrationality premise.

T194: actual- π return seeds. Let $N_{194} = \text{Theory.PiDigits.T194CentralPiReturnSeed}$.

[STATUS: MACHINE-CHECKED] $N_{194}.\text{exists.central.pi.unitBlock.surplus}$: for every $k \geq 3$, there exist unprescribed $n, A \in \mathbb{N}$ such that $A < q_k$ and $\mathcal{S}(q_k, A, n) > \frac{3}{20} q_k$. There is no timing constraint beyond $n \in \mathbb{N}$. Apart from the displayed hypothesis $3 \leq k$, the signature has no external premise and, in particular, no irrationality-measure premise; its proof uses Mathlib's theorem `irrational.pi`. [A]

[STATUS: MACHINE-CHECKED] $N_{194}.\text{eventually.exists.central.pi.unitBlock.surplus.before.eight.scale}$ (premise H_8) and $N_{194}.\text{eventually.exists.central.pi.unitBlock.surplus.before.eight.scale.ofBelowNine}$ (premise H_9):

$$\exists k_0 \forall k, \max(3, k_0) \leq k \implies \exists n, A, \quad n < 8q_k, \quad A < q_k, \quad \mathcal{S}(q_k, A, n) > \frac{3}{20} q_k.$$

[A]

[STATUS: MACHINE-CHECKED] $N_{194}.\text{eventually.exists.central.pi.unitBlock.surplus.in.fresh.horizon}$ (premise H_8) and $N_{194}.\text{eventually.exists.central.pi.unitBlock.surplus.in.fresh.horizon.ofBelowNine}$ (premise H_9): the same onset quantifiers hold, with

$$q_k \leq n < 10q_k, \quad A < q_k, \quad \mathcal{S}(q_k, A, n) > \frac{3}{20} q_k.$$

[A]

[STATUS: MACHINE-CHECKED] $N_{194}.\text{eventually.exists.central.pi.unitBlock.surplus.with.predecessor.lift}$ (premise H_8) and $N_{194}.\text{eventually.exists.central.pi.unitBlock.surplus.with.predecessor.lift.ofBelowNine}$ (premise H_9): there is k_0 such that every $k \geq \max(3, k_0)$, with $q = q_k$, admits $n, A, d, C \in \mathbb{N}$ satisfying

$$\begin{aligned} q + 1 &\leq n < 10q, \quad A < q, \quad d < 10, \quad C < 10q, \\ A &= \lfloor q\xi_n \rfloor_{\mathbb{N}}, \quad d = \lfloor 10\xi_{n-1} \rfloor_{\mathbb{N}}, \quad C = A + dq = \lfloor 10q\xi_{n-1} \rfloor_{\mathbb{N}}, \\ \mathcal{S}(q, A, n) &> \frac{3}{20} q, \quad \mathcal{S}(10q, C, n-1) > \frac{3}{2} q. \end{aligned}$$

[A] The T194 public conclusions do not expose a witness y or an explicit central-coordinate equality. Such a witness, with $|y| \leq 9/22$, is constructed internally to invoke T193; likewise the predecessor proof internally constructs the same normalized coordinate y at both scales, with

$$\xi_n - c(q, A) = \frac{y}{q}, \quad \xi_{n-1} - c(10q, C) = \frac{y}{10q},$$

but its exported conclusion is exactly the floor/digit and surplus data displayed above. No coherent choice across scales, ray, recurrence, or positivity of an entire horizon is asserted. T198: exact Machin 3/7 brackets and MC0 equivalence. Let $N_{198} = \text{Theory.PiDigits.T198MachinBracketPack}$. [STATUS: MACHINE-CHECKED] With L_m and U_m as in Section 3, the theorem $N_{198}.\text{machin37_strict_bracket_and_width}$ says that every $m \in \mathbb{N}$ satisfies $L_m < \pi < U_m$ and the exact width formula (3.4). [A]
 [STATUS: MACHINE-CHECKED] $N_{198}.\text{tendsto_machin37_endpoints}$: both exact endpoint sequences tend to π . The declaration does not assert monotonicity. [A]
 [STATUS: MACHINE-CHECKED] $N_{198}.\text{piZeroBlock_iff_decimalCylinder}$: for every $n, \ell \in \mathbb{N}$ with $1 \leq \ell$, the next ℓ canonical `piDigits` from zero-based position n are all zero if and only if π lies in a half-open interval $[k/10^n, k/10^n + 10^{-(n+\ell)})$ for some $k \in \mathbb{Z}$. [A]
 [STATUS: MACHINE-CHECKED] $N_{198}.\text{machinMC0_iff_piZeroCylinder}$: the all- ℓ bracket-containment event (3.5) is equivalent to the corresponding all- ℓ decimal-cylinder event for π . [A]
 [STATUS: MACHINE-CHECKED] $N_{198}.\text{machinMC0_iff_piCW0}$: the bracket event MC0 is equivalent to occurrence of every positive-length all-zero word in the canonical `piDigit` sequence, which is exactly (3.6). [A] Neither MC0 nor CW0 is proved.
 T199: exact base-10 BBP shadow and endpoint equivalences. Let $N_{199} = \text{Theory.PiDigits.T199BBPShadowPack}$.
 [STATUS: MACHINE-CHECKED] $N_{199}.\text{bbp10_defect_bounds_and_exact_shadow}$: for every $n \geq 2$, with B_n, E_n, v_n, a as in Section 2 and $h_n = \text{fract}((10^n - 16)B_n)$, one has

$$0 < E_n < (5/8)^n, \quad v_n = \text{fract}(h_n + E_n),$$

the exact integer-plus-51 decimal lift used in (2.19), and the circle-norm bound (2.7). [A]
 [STATUS: MACHINE-CHECKED] $N_{199}.\text{bbp10_theta_formula_strictAnti_tendsto}$: the exact formula (2.11) holds for every $n \geq 2$, $\Theta_{n+1} < \Theta_n$, $\Theta_n \rightarrow 144\pi$, and $\Theta_n - 144\pi = 10E_n - E_{n+1}$. [A]
 [STATUS: MACHINE-CHECKED] $N_{199}.\text{bbp10_affine_fixedPoint}$: the affine recursion, the values $c = 144\pi - 452$ and $a = 51 - 16\pi \in (0, 1)$, the carry-7 fixed-point identity, $\text{fract}(16\pi) = 1 - a$, and $v_n = \text{fract}(x_n + a)$ are exactly (2.13)–(2.16). [A]
 [STATUS: MACHINE-CHECKED] $N_{199}.\text{bbp10_rightApproach_iff_piCW0}$: approach of v_n to a from the right through every positive neighborhood is equivalent to occurrence of every positive-length all-zero word in the canonical `piDigit` sequence, the CW0 half of (2.17). [A]
 [STATUS: MACHINE-CHECKED] $N_{199}.\text{bbp10_leftApproach_iff_piCW9}$: approach of v_n to a from the left through every positive neighborhood is equivalent to occurrence of every positive-length all-nine word in the canonical `piDigit` sequence, the CW9 half of (2.17). [A]
 [STATUS: MACHINE-CHECKED] $N_{199}.\text{bbp10_soh0_iff_piCW0}$: the arbitrary-lower-bound positive residue-plus-defect condition $\text{SOH}_{\text{BBP},10}^0$ is equivalent to occurrence of every positive-length all-zero word in the canonical `piDigit` sequence. [A] Neither side of any of these three endpoint equivalences is proved.

The V1 statement. [STATUS: MACHINE-CHECKED] Here the status applies only to the literal definitions in this row, not to the truth of V1. The exact definitions are `Theory.PiDigits.piDigit` and `Theory.PiDigits.V1`. Lean defines

$$\text{piDigit}(n) = \lfloor \pi 10^{n+1} \rfloor_{\mathbb{N}} \bmod 10 \quad \text{as an element of } \text{Fin } 10,$$

and

$$\text{V1} \iff \forall s : \text{List}(\text{Fin } 10), \exists n \in \mathbb{N}, \forall i < \text{length}(s), \quad \text{piDigit}(n + i) = s[i].$$

Thus V1 is the proposition that every finite decimal word occurs contiguously in the post-decimal digits of π . It is a *definition*, not a proved theorem; it takes no premise and is not registered by `audit/AxiomAudit.lean`. Nothing in `TheoryLib/PiDigits/T7Statements.lean` proves V1.

What is not formalized. The listed Lean interface does not formalize a proof of any published irrationality-measure bound for π , including the bounds attributed to Hata, Salikhov, or Zeilberger–Zudilin; H_8 and H_9 remain explicit assumptions. It does not formalize Theorems A–E or Theorem B' of [A], or their separator constructions [21]. Nor does it formalize the first equivalence in the Machin chain or the other displayed CW0 equivalence

$$\text{SOH}_{3/7} \iff \text{MC0}, \quad \text{P3} \iff \text{CW0}.$$

Those statements are proved in Sections 3 and 7, not as Lean declarations or Lean proofs. The remaining Machin equivalence $\text{MC0} \iff \text{CW0}$ is machine-checked by T198 as recorded above. Other Lean files do prove different equivalences involving V1—notably `Theory.PiDigits.T20.v1_iff_pi_baseTenOrbitDense` and, under H_8 , `Theory.PiDigits.MachinV1Equivalence.canonicalV1_iff_all_machinBlockCells_recurrent`—but neither is one of the two CW0 equivalences just listed. Finally, `TheoryLib/PiDigits/T7Statements.lean` contains the digit definition and the propositions V1–V3, but no proofs; this appendix claims machine checking there only for those literal definitions.

A.1. Ordinary mathematical bridge. For a real x , write $Tx = \{10x\}$. The following reconstruction records the ordinary decimal-orbit interface used around the declarations above.

Put

$$q = 10^k, \quad A_k(n, x) = \lfloor qT^n x \rfloor, \quad y_k(n, x) = qT^n x - A_k(n, x) - \frac{1}{2}.$$

Then exactly

$$(A.1) \quad y_k(n, x) = T^{n+k} x - \frac{1}{2}.$$

Thus

$$(A.2) \quad |y_k(n, x)| \leq \frac{9}{22} \iff T^{n+k} x \in J := \left[\frac{1}{11}, \frac{10}{11} \right].$$

A next digit in $\{1, \dots, 8\}$ places the orbit in

$$(A.3) \quad \left[\frac{1}{10}, \frac{9}{10} \right) \subset J.$$

The interval is half-open, as required by the canonical decimal convention.

A.1.1. Untimed central return. Every irrational x has arbitrarily late visits to J . Indeed, suppose that after time L the orbit avoids J . It must remain in one of

$$[0, 1/11) \quad \text{or} \quad (10/11, 1).$$

In the lower component, avoidance gives

$$T^{L+r} x = 10^r T^L x < 1/11 \quad (r \geq 0),$$

forcing $T^L x = 0$, hence $x \in \mathbb{Q}$. In the upper component,

$$1 - T^{L+r} x = 10^r (1 - T^L x) < 1/11,$$

which forces the upper trapping limit to be the circle endpoint $1 = 0$, hence rational; 1 is not literally a point of $[0, 1)$. This proves the claim.

A.1.2. *Timed return from IrrationalityMeasureBelow x 8.* Suppose

$$\text{IrrationalityMeasureBelow}(x, 8).$$

Choose a witnessing $\mu < 8$ and take $\varepsilon = 8 - \mu$. Then for all sufficiently large Q ,

$$(A.4) \quad Q^{-8} < \left| x - \frac{p}{Q} \right| \quad (p \in \mathbb{Z}).$$

If the orbit avoids J for every $m \in [s, 8s]$, the trapping argument gives, with $Q = 10^s$, either

$$\left| x - \frac{\lfloor Qx \rfloor}{Q} \right| < \frac{1}{11} 10^{-8s} < Q^{-8}$$

or the analogous inequality with numerator $\lfloor Qx \rfloor + 1$, contradicting (A.4).

Hence for all large s there is $m \in [s, 8s]$ with $T^m x \in J$. Take

$$s = 10^k + 1 + k, \quad n = m - k.$$

Then, for $k \geq 3$,

$$(A.5) \quad 10^k + 1 \leq n < 10^{k+1}.$$

Bad approximability implies the exact Lean-style exponent premise by taking $\mu = 2$: if

$$\left| x - \frac{p}{q} \right| \geq \frac{c}{q^2},$$

then for every $\varepsilon > 0$, sufficiently large q satisfy

$$(A.6) \quad q^{-(2+\varepsilon)} < cq^{-2} \leq \left| x - \frac{p}{q} \right|.$$

A.1.3. *Predecessor-coordinate predicate.* A *central witness* at level k is a tuple (n, A, y) satisfying

$$(A.7) \quad \begin{aligned} 10^k + 1 &\leq n < 10^{k+1}, & A &< q, & A &= \lfloor qT^n x \rfloor, \\ y &= qT^n x - A - \frac{1}{2}, & |y| &\leq \frac{9}{22}. \end{aligned}$$

The predecessor-coordinate predicate explicitly quantifies y :

$$(A.8) \quad \begin{aligned} P_6(x) : \quad \exists k_0 \forall k \geq \max(3, k_0) \exists n, A, d, C, y : \quad & q + 1 \leq n < 10q, \\ & A < q, \ d < 10, \ C < 10q, \\ & A = \lfloor qT^n x \rfloor, \\ & d = \lfloor 10T^{n-1} x \rfloor, \\ & C = A + dq = \lfloor 10qT^{n-1} x \rfloor, \\ & y = qT^n x - A - \frac{1}{2}, \\ & T^n x - \frac{A + 1/2}{q} = \frac{y}{q}, \\ & T^{n-1} x - \frac{C + 1/2}{10q} = \frac{y}{10q}, \\ & |y| \leq \frac{9}{22}. \end{aligned}$$

This is a closed proposition with y explicitly quantified.

To prove the identities, set

$$d = \lfloor 10T^{n-1} x \rfloor.$$

Then

$$T^n x = 10T^{n-1} x - d,$$

so

$$10qT^{n-1}x = dq + qT^n x.$$

Taking floors gives

$$\lfloor 10qT^{n-1}x \rfloor = dq + A = C,$$

and subtracting $C + \frac{1}{2}$ gives

$$(A.9) \quad 10qT^{n-1}x - C - \frac{1}{2} = qT^n x - A - \frac{1}{2} = y.$$

Thus predecessor lifting is an identity once a timed central witness has been chosen.

A.1.4. *Infinite omega-limit set: the circle correction.* Work on

$$\mathbb{T} = \mathbb{R}/\mathbb{Z}$$

with the circle metric, and let $Tz = 10z$ on $\mathbb{T}$. This is a continuous map of a compact metric space. Define

$$\omega_T(x) = \bigcap_{N \geq 0} \overline{\{T^n x : n \geq N\}}^{\mathbb{T}}.$$

For a continuous map on a compact metric space,

$$T(\omega_T(x)) = \omega_T(x).$$

For the nontrivial surjectivity inclusion, take any sequence $T^{n_j}x \rightarrow z \in \omega_T(x)$ with $n_j \rightarrow \infty$. Compactness gives a convergent subsequence of the predecessors $T^{n_j-1}x \rightarrow y \in \omega_T(x)$, and continuity gives $T(y) = z$.

If $\omega_T(x)$ were finite, T would therefore permute it. For some $p \geq 1$, T^p would fix every point of $\omega_T(x)$. Since the orbit approaches its omega-limit set,

$$(A.10) \quad d_{\mathbb{T}}(T^{n+p}x, T^n x) \rightarrow 0.$$

For

$$u = (10^p - 1)x$$

this says

$$\|10^n u\|_{\mathbb{T}} \rightarrow 0.$$

The trapping argument above then forces $u \in \mathbb{Q}/\mathbb{Z}$, hence $x \in \mathbb{Q}/\mathbb{Z}$, a contradiction. Therefore every irrational x has infinitely many circle omega-limit points.

A.1.5. *Free-phase inequalities.* Define at paper level a free-phase primitive atom

$$\mathfrak{p}_{q,A}(z)$$

by replacing `piOrbit n` in the formula for `primitiveBoundaryAtom` by $z \in [0, 1)$. The T192–T193 estimates use the phase only pointwise and through the central-coordinate equality. Replacing `piOrbit n` by z therefore gives the routine ordinary-mathematics generalization: if

$$z - \frac{A + 1/2}{q} = \frac{y}{q}, \quad |y| \leq \frac{9}{22},$$

then

$$(A.11) \quad \Re \text{boundaryMinorant}(q, y/q) > \frac{4859}{10000},$$

$$(A.12) \quad \Re \mathfrak{p}_{q,A}(z) > \frac{7139}{45000},$$

and

$$(A.13) \quad q \Re \mathfrak{p}_{q,A}(z) - \frac{7}{3q} > \frac{3q}{20}.$$

The numerical combination is

$$\frac{9}{20} \frac{4859}{10000} - \frac{108019}{1800000} = \frac{7139}{45000},$$

and

$$\frac{7139}{45000} - \frac{3}{20} = \frac{389}{45000}.$$

For $q \geq 1000$,

$$\frac{7}{3q} < \frac{389}{45000}q.$$

The declared specialization has $q \geq 1000$ because $q = 10^k$ with $k \geq 3$. For this isolated inequality alone,

$$q^2 > \frac{105000}{389} \approx 269.92,$$

so integer $q \geq 17$ already suffices.

At the predecessor, (A.9) supplies the same y at scale $10q$, yielding

$$(A.14) \quad U_{10q,C}^x(n-1) > \frac{3(10q)}{20} = \frac{3q}{2}.$$

The generic T192–T193 component inequalities are among the declarations itemized above. The combined paper-level reconstruction and its predecessor application are ordinary mathematical packaging, not an additional Lean declaration.

APPENDIX B. STABLE REPRODUCIBILITY RECORD

The machine-checked inventory in Appendix A is pinned to the PI repository commit `c9e4186ae005cf7242113be7f17bc40cd189e2bd`. The toolchain is Lean `leanprover/lean4:v4.30.0`, pinned by `lean-toolchain`, with Mathlib `v4.30.0`, pinned by `lakefile.toml`.

After checking out that commit in the PI repository, run the single verification command

```
pwsh workflows/verification/check.ps1
```

It builds the tracked Lean sources, applies the forbidden-shortcut scanner, compiles `audit/AxiomAudit.lean`, and enforces the axiom allowlist stated in Appendix A.

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